

Categories for the Working Mathematician - Self Study

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Abstract

This text contains my notes on Categories for the Working Mathematician, by Saunders Mac Lane. I read this book independently, starting in mid spring semester of my sophomore year, 2025. My notes largely paraphrase the text, and I work out whatever examples I see fit.

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Chapter 1

Categories, Functors, and Natural Transformations

1.1 Axioms for Categories

We start by describing categories using *metagraphs*, which do not employ the use of set theory. A metagraph consists of objects, arrows, and two operations:

- Domain, which maps arrows to an object $\text{dom } f = a$;
- Codomain, which maps arrows to an object $\text{cod } f = b$.

We write $f : a \rightarrow b$ to indicate f is an arrow with $\text{dom } f = a$ and $\text{cod } f = b$.

A *metacategory* is a metagraph together with two additional operations:

- Identity, assigning each item a an arrow 1_a ;
- Composite, assigning each pair (f, g) of arrows with $\text{dom } g = \text{cod } f$, an arrow $g \circ f : \text{dom } f \rightarrow \text{cod } g$ called their composite;

such that composition is associative and the identity arrows 1_a are identities with respect to composition. Alternative notations for the identity 1_a include id_a , and simply a itself. Examples of metacategories include the usual category **Set** of sets and **Grp** of groups, topological spaces with continuous maps as arrows, and compact Hausdorff spaces with the same as arrows.

Since identity arrows are uniquely determined for each object, it's possible to dispense entirely of the objects in a metacategory and define an arrows-only metacategory C . There exist a collection of arrows, for which some pairs (f, g) of arrows are called *composable*. For any composable arrows, we say the composition $g \circ f$ is defined, and we can then define an identity arrow to be any arrow u satisfying $f \circ u = f$ and $u \circ g = u$ for any f, g such that the composition is defined. The data must then satisfy the axioms

- $(k \circ g) \circ f$ is defined iff $k \circ (g \circ f)$ is defined, and they are equal;
- The composite kgf is defined if and only if the composites kg and gf are defined;
- For each arrow g of C , there exist identity arrows u, u' such that gu and $u'g$ are defined.

The final axiom indeed shows that identity arrows are unique in this regard, and allows defining of a domain and codomain for the arrow g . An arrows-only metacategory thus satisfies the usual metacategory axioms, if identity arrows are taken as objects.

1.2 Categories

Definition 1.1: Directed Graph

A directed graph is a set O of objects, a set A of arrows, and two functions

$$A \begin{matrix} \xrightarrow{\text{dom}} \\ \xrightarrow{\text{cod}} \end{matrix} O$$

where the set of composable pairs is the set

$$A \times_O A = \{(g, f) \mid g, f \in A, \text{dom } g = \text{cod } f\}.$$

Definition 1.2: Category

A category is a directed graph with two additional functions

$$O \xrightarrow{\text{id}} A, \quad A \times_O A \xrightarrow{\circ} A,$$

$$c \longmapsto 1_c, \quad (g, f) \longmapsto g \circ f,$$

where $g \circ f$ is usually written gf , such that

$$\text{dom } 1_a = \text{cod } 1_a, \quad \text{dom } gf = \text{dom } f, \quad \text{cod } gf = \text{cod } g.$$

for all $a \in O$ and $(g, f) \in A \times_O A$, such that a metacategory is formed.

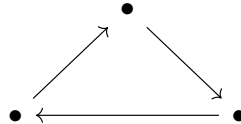
That is, a category is an interpretation of a metacategory within set theory. If C is a category, we usually write $a \in C$ or $f \in C$ to say a is an object of C , or f is an arrow of C . We also write

$$\text{Hom}(b, c) = \{f \in C \mid \text{dom } f = b, \text{cod } f = c\}$$

for the set of arrows $b \rightarrow c$. We will show later that this definition of a category is the same as saying a category is a monoid for the product $\times_O$.

Examples of categories include

- **0**, the empty category;
- **1**, the category with one object and one arrow;
- **2**, the category with two objects and precisely one non-identity arrow $a \rightarrow b$;
- **3**, the category with three objects whose non-identity arrows form a triangle



- **↓**, the category with two objects and two non-identity arrows $a \rightrightarrows b$; we call such arrows *parallel arrows*.

A category is called *discrete* when all arrows are identities. Any set forms a discrete category, and any discrete category is completely determined by its arrows.

A *monoid* is a category with one object. A monoid is thus fully described by its arrows, and its composition operation. A monoid can thus be described as a set M (of arrows) together with a binary operation $M \times M \rightarrow M$ and an identity element; such an object is precisely a semigroup with unity. Thus, for any category C and any object $a \in C$, the category $\text{Hom}(a, a)$ forms a monoid. In this vein of thought, a group is thus a category where every arrow has a two-sided inverse.

Given a set V of sets, the category $\mathbf{Ens}_V$ takes V as the objects, and morphisms as any regular functions between sets in V . $\mathbf{Ens}$ refers to an arbitrary category of this type.

An ordinal number n can be seen as the set $\{0, \dots, n-1\}$, and hence each ordinal is a category when viewed as a preorder. We can also consider the category ω , the first infinite cardinal, which when viewed as a category looks like

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \dots$$

The category Δ takes all finite ordinals as objects and all order-preserving functions $f : m \rightarrow n$ as arrows. This category is generally called the *simplicial category*, and it is extremely important.

We want an actual category $\mathbf{Set}$, which is clearly not a set itself. We will circumvent this by allowing there to exist a “universe” $\mathcal{U}$, and will call a set “small” if it belongs to the universe. We will construct this explicitly later (and I will be calling small sets “sets” and large sets “classes” like a normal fucking person). This allows us to construct

- $\mathbf{Set}$, objects are sets and morphisms are functions;
- $\mathbf{Set}_*$, objects are sets with a selected point, and morphisms are functions preserving these points;
- $\mathbf{Ens}$, category of all classes and functions within a variable set V ;
- $\mathbf{Cat}$, category of small categories and morphisms as functors;
- $\mathbf{Grp}$, category of groups and morphisms as group homomorphisms;
- $\mathbf{Ab}$, category of abelian groups;
- $\mathbf{Mon}$, category of (small) monoids;
- $\mathbf{Ring}$, category of rings;
- $\mathbf{CRing}$, category of commutative rings;
- $R\text{-}\mathbf{Mod}$, category of left R -modules;
- $\mathbf{Mod}\text{-}R$, category of right R -modules;
- $\mathbf{Top}$, category of topological spaces;
- $\mathbf{Toph}$, category of topological spaces with morphisms as homotopy classes of maps;
- $\mathbf{Top}_*$, category of topological spaces with a distinguished base point; base preserving continuous maps.

1.3 Functors

Definition 1.3: Functor

Let C, D be categories. A functor $T : C \rightarrow D$ consists of an object function T mapping each $c \in C$ to some $Tc \in D$, and an arrow function mapping each $f : c \rightarrow c'$ in C to $Tf : Tc \rightarrow Tc'$ in D , in such a way that

$$T1_c = 1_{Tc}, \quad T(g \circ f) = T(g) \circ T(f)$$

whenever the composite $g \circ f$ is defined in C .

For example, the power set functor $\mathcal{P} : \mathbf{Set} \rightarrow \mathbf{Set}$, defined such that for any $f : A \rightarrow B$, the new map $\mathcal{P}f : \mathcal{P}A \rightarrow \mathcal{P}B$ is such that for any $S \subseteq A$, $\mathcal{P}f(S) = f(S)$.

Functors first arose in algebraic topology. For any n , a topological space X can be assigned an abelian group $H_n(X)$ called the n -th homology group of X . However, for any $X \rightarrow Y$, there exists an induced $H_n(f) : H_n(X) \rightarrow H_n(Y)$ morphism of groups, and thus H_n becomes a functor $\mathbf{Top} \rightarrow \mathbf{Ab}$. We can in fact show homology groups are invariant under homotopy, and so we can equally define a functor $\mathbf{Top} \rightarrow \mathbf{Grp}$. The homotopy groups π_n are also functors; since they depend on a base point they define a functor $\mathbf{Top}_* \rightarrow \mathbf{Grp}$.

A functor which “forgets” underlying structure of an object is called a *forgetful functor*. For example, there is an apparent forgetful functor $U : \mathbf{Grp} \rightarrow \mathbf{Set}$, another $U : \mathbf{Ring} \rightarrow \mathbf{Ab}$, and so on.

Clearly functors may be composed, and moreover there is an identity functor 1_C for any category C . We may thus consider the metacategory of all categories, or the category $\mathbf{Cat}$ of all small categories.

A functor is called an *isomorphism* if it is two-sided invertible; in general isomorphism is too strong a condition, and weaker conditions are more useful. A functor is *full* if for every $c, c' \in C$ and every arrow $g : Tc \rightarrow Tc'$ in D , there exists $f : c \rightarrow c'$ in C such that $g = Tf$. A functor is *faithful*, or an embedding, if for all $c_1, c_2 \in C$ and parallel arrows $f_1, f_2 : c_1 \rightarrow c_2$ such that $Tf_1 = Tf_2$, we have $f_1 = f_2$.

These properties can be realized more intuitively. A functor defines functions $T : \text{Hom}(c, c') \rightarrow \text{Hom}(Tc, Tc')$. A functor is full when these functions are surjective; it is faithful when they are injective. A fully faithful functor is then a functor where these functions are bijections.

A subcategory S of a category C is a subset of the objects and arrows in C in such a way that S forms a category. There is automatically a faithful inclusion functor $S \hookrightarrow C$; we say a subcategory is *full* when the inclusion functor is full. Intuitively, S contains all arrows in C for the specified subcollection of objects.

1.4 Natural Transformations

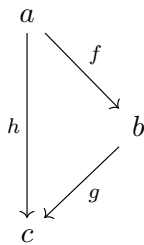
Definition 1.4: Natural Transformation

Let $T, S : C \rightarrow D$ be functors. A natural transformation $\tau : S \Rightarrow T$ is a function which assigns to each $c \in C$ an arrow $\tau_c = \tau c : Sc \rightarrow Tc$ of D , called the *component of τ at c* , such that for all $f : c \rightarrow c'$ in C , the diagram

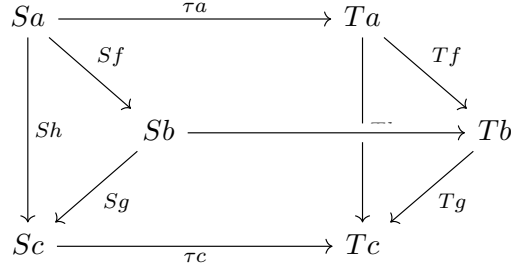
$$\begin{array}{ccc} Sc & \xrightarrow{\tau c} & Tc \\ Sf \downarrow & & \downarrow Tf \\ Sc' & \xrightarrow{\tau c'} & Tc' \end{array}$$

commutes.

We say any component $\tau c : Sc \rightarrow Tc$ is *natural* in c . We can think of a natural transformation $\tau : S \Rightarrow T$ as a way of transforming the image of a functor S into that of a functor T . For example, given a triangle



we would expect the diagram



commute. Indeed, it does.

Natural transformations define *morphisms of functors*. If a natural transformation has every component invertible, we call it a *natural isomorphism* and write $S \cong T$.

For example, the determinant is a natural transformation $\det_K : \mathbf{GL}_n \Rightarrow ()^*$, where $K \mapsto K^*$ maps a ring to its group of units. These are clearly two functors, both maps $\mathbf{CRing} \rightarrow \mathbf{Grp}$, and it's not that difficult to see.

Definition 1.5: Equivalent of Categories

An equivalence of categories $\mathcal{C}, \mathcal{D}$ is a pair of functors $F : \mathcal{C} \rightarrow \mathcal{D}$ and $G : \mathcal{D} \rightarrow \mathcal{C}$, which are inverses up to natural isomorphism; meaning there exist natural isomorphisms

$$1_{\mathcal{C}} \cong G \circ F, \quad 1_{\mathcal{D}} \cong F \circ G.$$

1.5 Monics, Epis, and Zeros

This section formalizes some familiar notions category-theoretically. Isomorphisms are defined as invertible arrows.

Definition 1.6: Monic/Epi

A morphism $f : a \rightarrow b$ is monic if for all morphisms $g_1, g_2 : z \rightarrow a$, if $f \circ g_1 = f \circ g_2$, then $g_1 = g_2$. In other words, monic morphisms (monomorphisms) are left-cancellable. Similarly, epimorphisms are right-cancellable; if $f : a \rightarrow b$ and $g_1, g_2 : b \rightarrow z$, then f is epi iff $g_1 \circ f = g_2 \circ f$ implies $g_1 = g_2$.

In **Set**, monomorphisms are precisely injections, and epimorphisms are precisely surjections. Monics and epis need not have left/right inverses, but if a morphism has a left/right inverse, it is necessarily monic/epi.

Definition 1.7: Section/Retraction

A right-inverse of a morphism f is a morphism g such that $gf = 1$, and a left-inverse is a morphism h such that $fh = 1$. Right-inverses of f are called *sections* of f , and left-inverses are called *retractions* of f . If f has a left-inverse, we say it is a *split* monic, and if it has a right inverse, it is a *split* epi.

Idempotent arrows are endomorphisms f such that $f^2 = f$, and split idempotent morphisms are idempotent morphisms f such that $f = hg$, and $gh = 1$.

Definition 1.8: Initial/Terminal

An initial object I is an object such that there is a unique arrow $I \rightarrow x$ for all objects x , and a terminal object T is an object such that there is a unique arrow $x \rightarrow T$ for all x . An object that is initial and terminal is called a null object or a 0 object, and the unique arrows are called zero-morphisms.

Any two initial objects are easily isomorphic, and the same with terminal objects. Any composition with a zero morphism is easily a zero morphism.

Definition 1.9: Groupoid

A groupoid is a category where every morphism is an isomorphism. A group is then a groupoid with one element.

A typical example of a groupoid is the fundamental groupoid $\pi(X)$ of a topological space X . Objects are points in X , and morphisms are homotopy classes of paths between points. Since every arrow is invertible, endomorphism sets $\text{End}_G(x)$ are groups for any groupoid G . The first fundamental group $\pi_1(X, x)$ can then be defined as $\text{End}_{\pi(X)}(x)$.

1.6 Foundations

Herein, we have assumed the existence of many classes, such as the class of all groups, all sets, etc. For example, consider the class of all groups. Let φ be the property that a set x is a group. Then the way we describe $\text{Obj}(\mathbf{Grp})$ is given by $\{x \mid \varphi(x)\}$. This amounts to the axiom of comprehension, which cannot be adopted in generality due to paradoxes such as Russel's paradox, $\{x \mid x \notin x\}$. ZFC assumes the axiom of restricted comprehension, which requires that the set-builder notation for constructing sets be defined as a subset of some other set, such as $\{x \in X \mid \varphi(x)\}$. However, there can be no set of all sets, for similar reasons, so this makes it impossible to construct $\text{Obj}(\mathbf{Set})$, or $\text{Obj}(\mathbf{Grp})$.

To avoid these paradoxes, we assume in addition to ZFC the existence of a *universe*, sometimes known as a Grothendieck universe.

Definition 1.10: Universe

A universe U is a set such that

1. $x \in u \in U$ implies that $x \in U$;
2. $u, v \in U$ implies that $\{u, v\} \in U$, $(u, v) \in U$, and $u \times v \in U$;
3. $x \in U$ implies that $\mathcal{P}(x) \in U$ and $\cup x \in U$ ($\cup x$ is the union over all elements of x);
4. $\omega \in U$ for ω the set of all finite ordinals;
5. if $f : a \rightarrow b$ is surjective with $a \in U$, then $b \in U$.

Some of these properties are clearly redundant.

These properties ensure that U remains closed under the standard operations of set theory, and thus we can safely perform set theory within U , and thus, all of ordinary mathematics can be performed safely within U , since $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$ can all be derived from $\omega \cong \mathbb{N}$.

This means that we can only consider sets within the universe when performing math, and thus simply write $\text{Obj}(\mathbf{Grp}) = \{x \in U \mid \varphi(x)\}$. Usually we will not write $x \in U$, since it becomes repetitive. Moreover, we choose $\text{Obj}(\mathbf{Set}) = U$. In other words, we think as if the only sets that exist are elements of U .

This leads us to some terminology. Small sets are sets that are contained within U ; i.e. x is a small set iff $x \in U$; and large sets are sets that are not elements of U . This includes U itself; we often call large sets *proper classes*. We use the word *class* to denote any subset of U , and all small sets are easily shown to be classes. Functions between small sets can also be regarded as small sets; we call them *small functions*. For my notes, I reserve the word “set” to mean exclusively “small set”, and will write “proper class” instead of “large set” when possible, since this appears to be more modern terminology.

We can then similarly append the word “small” and “large” to various other constructs, considering for example *small groups* and *small rings*, although I will simply call these structures by their standard

name, unless a distinction needs to be made. Distinctions are usually made for categories, since it is often important to discuss when a category is small vs when one is large. Note that $U \notin U$, so **Set** is not a small category, as this would contradict the axiom of regularity. Similarly, it can be shown that **Grp** is not small.

It seems prudent at this point to make precise what list of axioms we are using. The axioms of ZFC on a membership relation $\in$ guarantee

- the set $\emptyset$ exists;
- for any sets u, v , the sets $\{u, v\}, u \times v, \mathcal{P}(u), \cup u, (u, v)$ exist;
- sets with the same elements are equal (Extensionality);
- there exists an inductive set (Infinity);
- choice functions exist (Choice)
- there are no infinite chains $\cdots x_n \in \cdots \in x_0$, and no set contains itself (Regularity);
- the image of a function is a set (Replacement).

We will on occasion need to consider large sets which are not classes, such as the category **Cls** of all classes, defined as $\text{Obj}(\mathbf{Cls}) = \mathcal{P}(U)$. We will make distinction between classes and large sets at such a time, if necessary.

This allows discussion of all categories, but not all metacategories. Attempting to describe all of these leads to some strange issues, and as such there has been discussion as to whether set theory is the proper foundational device for category theory, and in some cases, mathematics itself.

1.7 Large Categories

We just list a lot of large categories I am already familiar with, and their monics. Since I know all of them already, I will simply list the relevant information given. All of these are assumed to contain all objects of the given type which can be formed from sets.

- **Ab**: monics are injectives, epis are surjectives;
- **Ring**: monics are injectives, and surjectives are epis, but not the converse. $\mathbb{Z}$ is initial and 0 is terminal;
- **Top**: monics are injections and epis are surjections, the singleton is terminal and the empty space is initial;
- **Toph**: homotopy classes of injectives *need not be monics*;
- **Set_{*}**: the singleton is initial and terminal (zero object), monics are left-invertible maps, epis are right-invertible maps, and isomorphisms are precisely those maps that are monic and epi;
- **Top_{*}**: no information given as to the monics;
- **Rel**: the category of relations; $\text{Obj}(\mathbf{Rel}) = U$, and morphisms are given by relations:

$$\text{Hom}(X, Y) = \{R \subseteq X \times Y\}.$$

Composition $S \circ R$ for $S : Y \rightarrow Z$ and $R : X \rightarrow Y$ is given by

$$S \circ R = \{(x, z) \mid (x, y) \in R \wedge (y, z) \in S\}.$$

The identity arrow is the identity relation, and **Rel** holds **Set** as a subcategory.

Definition 1.11: Concrete Category

A *concrete category* is a pair (C, U) with C a category and U a forgetful functor $U : C \rightarrow \mathbf{Set}$. This is understood intuitively as being a category where each object has an underlying set, and morphisms are functions between these sets.

All of the above categories, except **Toph** and **Rel**, are concrete categories. Many categories we will encounter are concrete, but many are not. This makes our notion of a category more “abstract” than that of a concrete category.

1.8 Hom-Sets

Recall that $\text{Hom}_C(x, y) = \{f \in \text{Mor}(C) \mid f : x \rightarrow y\}$. This is often also written as $\text{hom}_C(x, y)$, $C(x, y)$, or $\text{Hom}(x, y)$. These are called the hom-sets, and as we have seen before, categories can be defined entirely via their hom-sets. In this definition (of small categories), composition is given by functions $\circ : \text{Hom}(b, c) \times \text{Hom}(a, b) \rightarrow \text{Hom}(a, c)$ for each triple (a, b, c) . Associativity is then described by the commutative diagram

$$\begin{array}{ccc} \text{Hom}(c, d) \times \text{Hom}(b, c) \times \text{Hom}(a, b) & \longrightarrow & \text{Hom}(b, d) \times \text{Hom}(a, b) \\ \downarrow & & \downarrow \\ \text{Hom}(c, d) \times \text{Hom}(a, c) & \longrightarrow & \text{Hom}(a, d) \end{array}$$

Functors and natural transformations can be formulated similarly.

An important thing to note is that oftentimes the hom-sets can have additional structure. One simple case (that I’ve already seen) is where hom-sets are abelian groups; we call such a category an **Ab**-category or a *preadditive* category. In these categories, hom-sets come with an abelian addition operation, and composition is bilinear with respect to addition, meaning

$$f \circ (g + h) = (f \circ g) + (f \circ h),$$

and likewise for the other side.

Since composition in an **Ab**-category A is bilinear

$$A(b, c) \times A(a, b) \rightarrow A(a, c),$$

it can be written with a tensor product as a linear map

$$A(b, c) \otimes_{\mathbb{Z}} A(a, b) \rightarrow A(a, c).$$

We can thus describe an **Ab**-category as

- A collection of objects $\text{Obj}(A)$;
- A function which assigns pairs to abelian groups $(a, b) \mapsto A(a, b)$;
- For each triple (a, b, c) , a linear morphism of abelian groups $A(b, c) \otimes_{\mathbb{Z}} A(a, b) \rightarrow A(a, c)$;

such that the associative and unit laws are satisfied. The definition of a preadditive category is the same as that of a category, except $\times$ is replaced with $\otimes_{\mathbb{Z}}$, and the singleton is replaced with $\mathbb{Z}$ as the unit under tensor.

This evidently generalizes; given a category M with a kind of tensor product $\otimes$ and a unit I , we may be able to view hom-sets in a category A as objects in M . Such categories are called M -enriched categories, and we will see later how to formalize this notion.

Finally, additive functors are functors $F : A \rightarrow B$ which determine for each abelian group $A(a, b)$ and $B(F(a), F(b))$, an abelian group homomorphism $F : A(a, b) \rightarrow B(F(a), F(b))$. Denote by **Ab-cat** the category of all small **Ab**-categories, with morphisms as additive functors.

Chapter 2

Constructions on Categories

2.1 Duality

Duality is the process of reversing the arrows. In this section, we will describe duality axiomatically, independent of sets and functions.

The elementary theory of an abstract category (ETAC) consists of statements Σ with letters $a, b, c \dots$ for objects and letters $f, g, h \dots$ for arrows. *Atomic* statements consist of statements of the form

- a is the (co)domain of f ;
- i is the identity arrow of a ;
- $a = b$;
- $f = g$;
- $f = g \circ h$.

A statement Σ is any well-defined formula that can be constructed from atomic statements using standard logical operators and quantifiers. For example, the statement $f : a \rightarrow b$, which is a shorthand for “ a is the domain of f and b is the codomain of f ”, is a statement Σ .

Duals of statements in ETAC are formed by making the following (somewhat redundant) replacements:

Statement Σ	Dual statement Σ^*
$f : a \rightarrow b$	$f : b \rightarrow a$
$a = \text{dom } f$	$a = \text{cod } f$
$b = \text{cod } f$	$b = \text{dom } f$
$h = g \circ f$	$h = f \circ g$
f is monic	f is epi
u is a right-inverse of f	u is a left-inverse of f
t is terminal	t is initial

If a statement involves a diagram, the dual of that statement is the diagram given by reversing all arrows.

The dual of each axiom of a category is also an axiom, and thus replacing a proof with it's dual gives a valid proof for the dual statement. This is known as *duality*: given a statement Σ which follows from the axioms of a category, it's dual Σ^* also follows.

Duality also applies to statements about several categories and functors between them. This does *not* reverse the direction of the functor, it simply reverses the direction of the arrows in the categories.

2.2 Contravariance and Opposites

We consider the opposite category C^{op} , which flips all the arrows. It is easily shown a statement is true in C if and only if it's dual is true in C^{op} , and thus C^{op} can be viewed as a manifestation of the principle of duality.

We can also consider opposite functors; for any functor $F : C \rightarrow D$, the assignment $F^{\text{op}} : C^{\text{op}} \rightarrow D^{\text{op}}$ given by $x \mapsto F(x)$ and $f^{\text{op}} \mapsto F(f)^{\text{op}}$ is also clearly covariantly functorial. This means there is an assignment $C \mapsto C^{\text{op}}$ and $F \mapsto F^{\text{op}}$, which is in fact functorial, and defines a (covariant) functor $\mathbf{Cat} \rightarrow \mathbf{Cat}$.

We can also consider contravariant functors $G : C \rightarrow D$, where if $f : a \rightarrow b$ in C , then $G(f) : G(b) \rightarrow G(a)$. That is, contravariant functors flip the direction of arrows. They can equivalently be viewed as covariant functors $G : C^{\text{op}} \rightarrow D$, which this text and my notes will do throughout.

Hom-sets provide important examples of co- and contravariant functors. For each $a \in C$, the assignment $x \mapsto C(a, x)$ gives a covariant functor $C \rightarrow \mathbf{Set}$ by mapping morphisms as follows: for any $f : x \rightarrow y$, the induced morphism is given by

$$C(a, f) : C(a, x) \rightarrow C(a, y), \quad \varphi \mapsto f \circ \varphi.$$

This works since $\varphi : a \rightarrow x$, so $f \circ \varphi : a \rightarrow x \rightarrow y$. This is easily shown to be functorial.

Similarly, the contravariant hom-functor $x \mapsto C(x, a)$, written as $C(-, a) : C^{\text{op}} \rightarrow \mathbf{Set}$, has action on morphisms given by

$$C(f, a) : C(y, a) \rightarrow C(x, a), \quad \varphi \mapsto \varphi \circ f.$$

This works since $\varphi : y \rightarrow a$, so $\varphi \circ f : x \rightarrow y \rightarrow a$.

Another example is the *sheaf of germs of continuous functions on X* , for X a topological space. Let $\mathbf{Open}(X)$ be the poset category given by all open subsets of X under the partial order $\subseteq$. Let $\overline{C}(U)$ denote the set of all continuous functions $f : U \rightarrow \mathbb{R}$. The restriction assignment $f \mapsto f|_V$ is a function $\overline{C}(U) \rightarrow \overline{C}(V)$ for all $V \subseteq U$. This makes $\overline{C}$ a contravariant functor $\mathbf{Open}(X) \rightarrow \mathbf{Set}$.

Another example is the assignment $\mathbf{Ring} \rightarrow \mathbf{CAT}$ given by $R \mapsto \mathbf{Mod}\text{-}R$. For any ring homomorphism $\rho : R \rightarrow S$, the functor $\mathbf{Mod}\text{-}\rho$ is given by mapping S -modules M to R -modules M , with R -action $m \cdot r$ given by $m \cdot \rho(r)$.

2.3 Products of Categories

Definition 2.1: Product Category

Let C, D be categories. Define the *product category* $C \times D$ as the category where

$$\text{Obj}(C \times D) = \{(c, d) \mid c \in C, d \in D\},$$

and morphisms $(c_1, d_1) \rightarrow (c_2, d_2)$ are given by pairs (f, g) , where

$$f : c_1 \rightarrow c_2, \quad g : d_1 \rightarrow d_2$$

in C and D , respectively. Composition is defined pointwise.

Product categories come with projections $\pi_C : C \times D \rightarrow C$ and $\pi_D : C \times D \rightarrow D$ satisfying the universal property of products:

$$\begin{array}{ccccc} & & Z & & \\ & \swarrow F & \downarrow & \searrow G & \\ C & \xleftarrow{\pi_C} & C \times D & \xrightarrow{\pi_D} & D \end{array}$$

Two functors $U : C \rightarrow C'$ and $V : D \rightarrow D'$ also have a product $U \times V : C \times D \rightarrow C' \times D'$, given by

$$(U \times V)(x, y) = (U(x), V(y)); \quad (U \times V)(f, g) = (U(f), V(g)).$$

This is equivalently the unique arrow in the diagram

$$\begin{array}{ccccc} & & C \times D & & \\ & \swarrow U \circ \pi_C & \downarrow U \times V & \searrow V \circ \pi_D & \\ C' & \longleftarrow & C' \times D' & \longrightarrow & D' \end{array}$$

One clearly also has that you can compose products of functors pointwise, by a simple diagram chase, so product is infact a functor

$$\times : \mathbf{Cat} \times \mathbf{Cat} \rightarrow \mathbf{Cat}.$$

Similar functors exist $\mathbf{Grp} \times \mathbf{Grp} \rightarrow \mathbf{Grp}$, $\mathbf{Top} \times \mathbf{Top} \rightarrow \mathbf{Top}$, and many more. Functors from a product category to a regular category are called *bifunctors*. Bifunctors satisfy the property that if one of the arguments is held fixed and the other is allowed to vary, this induces a functor as well.

Proposition 2.1

Let B, C, D be categories. For all $b \in B$ and $c \in C$, let $L_c : B \rightarrow D$ and $M_b : C \rightarrow D$ be functors such that $M_b(c) = L_c(b)$ for all b and c . There exists a bifunctor $X : B \times C \rightarrow D$ with $S(-, c) = L_c$ and $S(b, -) = M_b$ for all b and c , if and only if for all $f : b \rightarrow b'$ and $g : c \rightarrow c'$, we have

$$M_{b'}(g) \circ L_c(f) = L_{c'}(f) \circ M_b(g).$$

These equal arrows are then the value $S(f, g)$.

Proof. First, let S be a bifunctor. Let $f : c \rightarrow \bar{c}$ and $g : b \rightarrow \bar{b}$. Then since $L_c(b) = M_b(c)$ for all pairs (b, c) , we have the equalities

$$M_b(c) = L_c(b), \quad L_c(\bar{b}) = M_{\bar{b}}(c), \quad M_b(\bar{c}) = L_{\bar{c}}(b), \quad M_{\bar{b}}(\bar{c}) = L_{\bar{c}}(\bar{b}).$$

Functoriality of M and L then gives us the commutative diagram

$$\begin{array}{ccccccc} L_c(b) & \xrightarrow{L_c(g)} & L_c(\bar{b}) = M_{\bar{b}}(c) & \xrightarrow{M_{\bar{b}}(f)} & M_{\bar{b}}(\bar{c}) & & \\ \parallel & & & & \parallel & & \\ M_b(c) & \xrightarrow{M_b(f)} & M_b(\bar{c}) = L_{\bar{c}}(b) & \xrightarrow{L_{\bar{c}}(g)} & L_{\bar{c}}(\bar{b}) & & \end{array}$$

This gives us the desired equality. The converse gives us this equality, allowing S to be well-defined, and functoriality is then manifest from functoriality of M and L . ■

Hom also gives an example of a bifunctor; if C is locally small, then $\mathbf{Hom}$ is a bifunctor $\mathbf{Hom} : C^{\text{op}} \times C \rightarrow \mathbf{Set}$. There is also an isomorphism $(B \times C)^{\text{op}} \cong B^{\text{op}} \times C^{\text{op}}$.

Natural transformations between bifunctors $\alpha : S \Rightarrow S'$ are viewed as follows. α is a function which assigns to each pair $(b, c) \in B \times C$, an arrow $\alpha(b, c) : S(b, c) \rightarrow S'(b, c)$ (the component of the natural transformation at (b, c)) satisfying the naturality condition.

We call α *natural in b* for each $c \in C$ if the components define another natural transformation

$$\alpha(-, c) : S(-, c) \Rightarrow S'(-, c)$$

for each $c \in C$.

Proposition 2.2

Let $S, \bar{S} : B \times C \rightarrow D$ be bifunctors. A function α is a natural transformation $\alpha : S \Rightarrow \bar{S}$ if and only if $\alpha(b, c)$ is natural in b for each c , and natural in c for each b .

Proof. First, let $\alpha : S \Rightarrow \bar{S}$. Since this defines a naturality condition between *all* pairs (b, c) , simply fixing a given c and letting b vary still induces a naturality condition, since action on morphisms can be viewed as pairs $(f, 1_c)$. For the converse, I will denote by $\alpha(b, -)_c$ the component of $\alpha(b, -)$ at c , and likewise for $\alpha(-, c)$. Note that $\alpha(b, -)_c = \alpha(b, c) = \alpha(-, c)_b$. Let $(f, g) : (b, c) \rightarrow (\bar{b}, \bar{c})$. This gives us the commutative diagram

$$\begin{array}{ccccc}
 & & S(f, g) & & \\
 & \swarrow & & \searrow & \\
 S(b, c) & \xrightarrow{S(1_b, g)} & S(b, \bar{c}) & \xlongequal{\quad} & S(b, \bar{c}) & \xrightarrow{S(f, 1_{\bar{c}})} & S(\bar{b}, \bar{c}) \\
 \downarrow \alpha(b, -)_c & & \downarrow \alpha(b, -)_{\bar{c}} = \alpha(-, \bar{c})_b & & \downarrow \alpha(-, \bar{c})_{\bar{b}} & & \\
 \bar{S}(b, c) & \xrightarrow{\bar{S}(1_b, g)} & \bar{S}(b, \bar{c}) & \xlongequal{\quad} & \bar{S}(b, \bar{c}) & \xrightarrow{\bar{S}(f, 1_{\bar{c}})} & \bar{S}(\bar{b}, \bar{c}) \\
 & \searrow & & \swarrow & \\
 & & \bar{S}(f, g) & &
 \end{array}$$

By commutativity, we have that $\alpha : S \Rightarrow \bar{S}$. ■

2.4 Functor Categories

Let $F, G, H : C \rightarrow D$ be functors, and let $\sigma : F \Rightarrow G$, $\tau : G \Rightarrow H$ be natural transformations. We can compose natural transformations to obtain a natural transformation $\tau \cdot \sigma : F \Rightarrow H$, given by $(\tau \cdot \sigma)_c = \tau_c \circ \sigma_c$. Diagrammatically, this is given by the commutative diagram

$$\begin{array}{ccc}
 F(c) & \xrightarrow{F(f)} & F(d) \\
 \downarrow \sigma_c & & \downarrow \sigma_d \\
 G(c) & \xrightarrow{G(f)} & G(d) \\
 \downarrow \tau_c & & \downarrow \tau_d \\
 H(c) & \xrightarrow{H(f)} & H(d)
 \end{array}
 \quad \begin{array}{c} (\tau \cdot \sigma)_c \\ \\ (\tau \cdot \sigma)_d \end{array}$$

Composition in this manner is manifestly associative, and it also has as an identity the identity transformation $1_F : F \Rightarrow F$, where each component is an identity morphism. This allows us to define the functor category.

Definition 2.2: Functor Category

Let C, D be categories. The *functor category* $\text{Fun}(C, D)$ takes as objects all functors $C \rightarrow D$, and as morphisms natural transformations. We often write

$$\text{Nat}(F, G) = \text{Hom}_{\text{Fun}(C, D)}(F, G).$$

Hom-sets in this category need not be small, and moreover functor categories need not be classes. They can be large sets which are not classes.

2.5 The Category of All Categories

We have defined what we call *vertical composition* of natural transformations, where we compose them along functors. There is, however, another way to compose natural transformations, known as *horizontal composition*. Let A, B, C be categories, and let

$$\begin{array}{ccccc} A & \xrightarrow{F_1} & B & \xrightarrow{F_2} & C \\ & \Downarrow \sigma & & \Downarrow \tau & \\ A & \xrightarrow{G_1} & B & \xrightarrow{G_2} & C \end{array}$$

commute. That is, $F_1, G_1 : A \rightarrow B$ and $F_2, G_2 : B \rightarrow C$ are functors, and $\sigma : F_1 \Rightarrow G_1$, $\tau : F_2 \Rightarrow G_2$ are natural transformations. We want to compose these natural transformations.

To do this, first form the composite functors $F_2 \circ F_1$ and $G_2 \circ G_1$. From hereon out, using components of natural transformations as subscripts will become impractical, so I will swap to the book's function notation, where $\sigma(c)$ is the component of σ at c .

Note that $\sigma(c) : F_1(c) \rightarrow G_1(c)$, and thus

$$F_2(\sigma(c)) : (F_2 \circ F_1)(c) \rightarrow (F_2 \circ G_1)(c).$$

Similarly, we have

$$G_2(\sigma(c)) : (G_2 \circ F_1)(c) \rightarrow (G_2 \circ G_1)(c).$$

Since τ is a natural transformation $F_2 \Rightarrow G_2$, it has components at $F_1(c)$ and $G_1(c)$, and thus by the naturality condition the square

$$\begin{array}{ccc} (F_2 \circ F_1)(c) & \xrightarrow{\tau(F_1(c))} & (G_2 \circ F_1)(c) \\ F_2(\sigma(c)) \downarrow & & \downarrow G_2(\sigma(c)) \\ (F_2 \circ G_1)(c) & \xrightarrow{\tau(G_1(c))} & (G_2 \circ G_1)(c) \end{array}$$

commutes. Define *horizontal composition* as the diagonal of this square, meaning

$$(\tau \circ \sigma)(c) = G_2(\sigma(c)) \circ \tau(F_1(c)) = \tau(G_1(c)) \circ F_2(\sigma(c)).$$

To show naturality of $\tau \circ \sigma$, consider the commutative diagram

$$\begin{array}{ccccc} (F_2 \circ F_1)(c) & \xrightarrow{F_2(\sigma(c))} & (F_2 \circ G_1)(c) & \xrightarrow{\tau(G_1(c))} & (G_2 \circ G_1)(c) \\ (F_2 \circ F_1)(c) \downarrow & & (F_2 \circ G_1)(f) \downarrow & & (G_2 \circ G_1)(f) \downarrow \\ (F_2 \circ F_1)(d) & \xrightarrow{F_2(\sigma(d))} & (F_2 \circ G_1)(d) & \xrightarrow{\tau(G_1(d))} & (G_2 \circ G_1)(d) \end{array}$$

The right square commutes by naturality of τ , and the left square commutes by naturality of σ and functoriality of F_2 . Thus, the diagram commutes. Note that the horizontal composites are precisely the components $(\tau \circ \sigma)(c)$ and $(\tau \circ \sigma)(d)$. Therefore, $\tau \circ \sigma : F_2 \circ F_1 \Rightarrow G_2 \circ G_1$ is natural.

Horizontal composition is once again manifestly associative, and moreover the identity natural transformation on the identity functor is an identity for horizontal composition. The book at this point adopts the notation of using F not only to denote a functor, but also to denote the identity natural transformation 1_F . I will not use this notation.

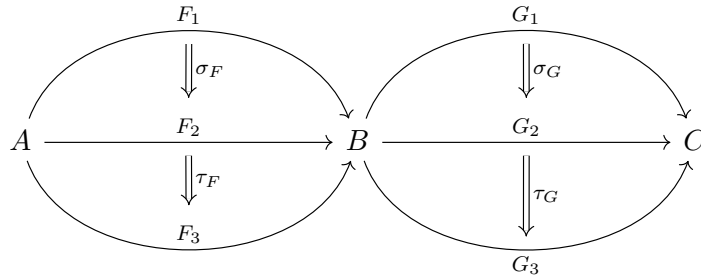
With notation as above, we also have that there are natural transformations

$$1_{F_2} \circ \sigma : F_2 \circ F_1 \Rightarrow F_2 \circ G_1, \quad \tau \circ 1_{G_1} : F_2 \circ G_1 \Rightarrow G_2 \circ G_1.$$

We can then redefine horizontal composition in generality as

$$\tau \circ \sigma = (1_{G_2} \circ \sigma) \cdot (\tau \circ 1_{F_1}) = (\tau \circ 1_{G_1}) \cdot (1_{F_2} \circ \sigma).$$

One may see by virtue of a short diagram chase (which I did on my whiteboard and do not wish to type) that given the situation



we have the equality

$$(\tau_G \cdot \sigma_G) \circ (\tau_F \cdot \sigma_F) = (\tau_G \circ \tau_F) \cdot (\sigma_G \circ \sigma_F).$$

This is known as the *interchange law*, which turns out to somewhat classify natural transformations (for small categories):

Theorem 2.1

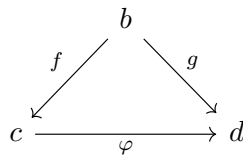
The collection of all natural transformations is the set of arrows of two different categories under two different operations of composition, $\cdot$ and $\circ$, which satisfy the interchange law. Moreover, any identity for $\circ$ is an identity for $\cdot$.

We often omit the $\circ$ when composing along categories, similarly to how we often omit $\circ$ when composing functors or morphisms. To make the distinction clear, we will generally retain the $\cdot$ for vertical composition.

A double category is a category which has arrows with two different compositions, composable along morphisms and objects, which satisfy the interchange law. A 2-category is a double category for which identities for $\circ$ are also identities for $\cdot$.

2.6 Comma Categories

Comma categories are categories where objects are certain kinds of arrows. For example, given an object $b \in C$, the category $(b \downarrow C)$ of objects *under* b , wherein objects are pairs (f, c) with $c \in C$ and $f : b \rightarrow c$; morphisms $(f, c) \rightarrow (g, d)$ are morphisms $\varphi : c \rightarrow d$ in C , such that the diagram



commutes.

A similar example is the category $(C \downarrow a)$ of objects *over* a , where objects are pairs (f, c) with $f : c \rightarrow a$, and morphisms $(f, c) \rightarrow (g, d)$ are commutative triangles

$$\begin{array}{ccc} c & \xrightarrow{\varphi} & d \\ & \searrow f \quad \swarrow g & \\ & a & \end{array}$$

in C . For example, $\{*\}$ is terminal in **Set**, so there is always a unique map $X \rightarrow \{*\}$, and thus $(\mathbf{Set} \downarrow \{*\})$ is isomorphic to **Set**.

Let $F : C \rightarrow D$ and $b \in D$. The category $(b \downarrow S)$ of objects S -under b , is formed of pairs (f, c) with $c \in C$ and $f : b \rightarrow F(c)$. Morphisms $(c, f) \rightarrow (d, g)$ in $(b \downarrow S)$ are morphisms $\varphi : c \rightarrow d$ in C , such that the diagram

$$\begin{array}{ccc} & b & \\ f \swarrow & & \searrow g \\ F(c) & \xrightarrow{F(\varphi)} & F(d) \end{array}$$

commutes. The category $(S \downarrow b)$ of objects S -over b is constructed analogously.

For example, let $U : \mathbf{Grp} \rightarrow \mathbf{Set}$ be the forgetful functor. For some $x \in \mathbf{Set}$, the category $(x \downarrow U)$ is given by functions $x \rightarrow U(g)$ for g a group; for example the function mapping x into the free group on x is one such object. Clearly this object is of some importance, and will be treated extensively later.

Here is the general construction of a comma category. Let A, B, C be categories, and let

$$A \xrightarrow{F} C \xleftarrow{G} B$$

be functors. The comma category $(F \downarrow G)$ takes triples (a, b, f) as objects with $a \in A$, $b \in B$, and $f : F(a) \rightarrow G(b)$. Arrows $(a_1, b_1, f_1) \rightarrow (a_2, b_2, f_2)$ are pairs (k, h) of arrows $k : a_1 \rightarrow a_2$ and $h : b_1 \rightarrow b_2$, such that

$$f_2 \circ F(k) = G(h) \circ f_1.$$

Equivalently, the diagram

$$\begin{array}{ccc} F(a_1) & \xrightarrow{F(k)} & F(a_2) \\ f_1 \downarrow & & \downarrow f_2 \\ G(b_1) & \xrightarrow{G(h)} & G(b_2) \end{array}$$

must commute. Composition of morphisms is done pointwise.

One recovers all of the previous examples by replacing objects b with functors $\mathbf{1} \rightarrow C$ mapping $1 \mapsto b$, and by replacing categories C with identity functors 1_C . Many categorical constructions can be recovered using comma categories. For example, we can consider $(a \downarrow b)$ for some $a, b \in C$, which takes as objects pairs (a, b, f) , where $f : a \rightarrow b$. Equivalently, $\text{Obj}(a \downarrow b) \cong \text{Hom}_C(a, b)$. Since $\mathbf{1}$ has no non-identity morphisms, there are no non-identity morphisms in this category, and it is thus a discrete category.

2.7 Graphs and Free Categories

Recall the construction of the free monoid $F(X)$ on the set X , which is formed by all finite strings in X , has multiplication given by juxtaposition, and for which any set-function $X \rightarrow M$ with M any monoid extends to a unique morphism of monoids $F(X) \rightarrow M$. To construct a free *category*, we replace a set with a directed graph.

Recall that a directed graph G is a pair (O, A) of sets of objects and arrows, together with functions $\text{dom}, \text{cod} : A \rightarrow O$. Morphisms $G \rightarrow G'$ are a pair of functions $D_O : O \rightarrow O'$ and $D_A : A \rightarrow A'$ such that

$$D_O \text{dom } f = \text{dom } D_A f, \quad D_O \text{cod } f = \text{cod } D_A f$$

for all $f \in A$. The category **Grph** takes as objects all small graphs, as morphisms all morphisms of graphs, and has a forgetful functor $U : \mathbf{Cat} \rightarrow \mathbf{Grph}$.

An O -graph is any graph for which O is the set of objects, and morphisms of O -graphs are morphisms of graphs for which D_O is the identity. We define the product over O of two O -graphs A, B to be

$$A \times_O B = \{(g, f) \mid \text{dom } g = \text{cod } f, g \in A, f \in B\};$$

the set of composable pairs. We then define $\text{dom}(g, f) = \text{dom } f$, and $\text{cod}(g, f) = \text{cod } g$. This product is associative, and the O -graph O , which takes dom and cod the identity functions, is a unit for $\times_O$. A category with $\text{Obj}(C) = O$ is then an O -graph together with functions c, i (composition and identity), satisfying the necessary coherence diagrams to ensure associativity of c and unity of i with respect to c .

In this sense, categories are similar to monoids, with $\times_O$ the product. O -graphs can then be used to generate free categories, similarly to monoids.

Theorem 2.2

Let $G = \{A \rightrightarrows O\}$ be a small graph. There is a small category $C = C_G$ with $\text{Obj}(C) = O$ and a morphism $P : G \rightarrow U(C)$ of graphs, for $U : \mathbf{Cat} \rightarrow \mathbf{Grph}$ the forgetful functor, such that for any category Z and any morphism $D : G \rightarrow U(Z)$ of graphs, there is a unique functor $D' : C \rightarrow Z$ making the diagram

$$\begin{array}{ccc} G & \xrightarrow{P} & C \\ & \searrow D & \downarrow D' \\ & & Z \end{array}$$

commute (with the obvious regards as to forgetful functors and the like). In particular, if $U(Z)$ is an O -graph, then D' is the identity on objects.

Proof. Note that the comma category $(G \downarrow U)$ takes as objects pairs (C, F) , where $F : G \rightarrow U(C)$ is a morphism of graphs, and C is a category. Morphisms $\varphi : (C, F_C) \rightarrow (D, F_D)$ are thus functors $\varphi : C \rightarrow D$ such that the diagram

$$\begin{array}{ccc} & G & \\ F_C \swarrow & & \searrow F_D \\ U(C) & \xrightarrow{U(\varphi)} & U(D) \end{array}$$

commutes. It thus follows that the free category C_G must be initial in the comma category $(G \downarrow U)$.

To define C , take $\text{Obj}(C) = O$ and arrows of C to be finite paths

$$a_1 \rightarrow a_2 \rightarrow \cdots \rightarrow a_{n-1} \rightarrow a_n.$$

Since composition is not necessarily defined in a graph, this is formalized as a pair $(a_1, f_1, \dots, a_n)$. We take each pair $(a_1, f_1, \dots, a_n)$ to be a morphism $a_1 \rightarrow a_n$ in C . Composition is defined via concatenation, meaning

$$(a_1, f_1, \dots, a_m) \circ (a_1, f_1, \dots, a_n) = (a_1, f_1, \dots, a_{n+m}) : a_1 \rightarrow a_{n+m}.$$

The morphism P of graphs simply maps arrows $f : a_1 \rightarrow a_2$ to the string (a_1, f, a_2) . Showing the existence of D' is fairly simple; the definition of D' is just $(a_1, f, a_2) \mapsto f$ for the smallest arrows, given by composition otherwise, and agrees with D on object mappings. The uniqueness and commutivity of the diagram is manifest/forced. ■

When O is a singleton, the graph reduces to a set X of arrows, and the theorem provides for us the construction of a free monoid.

Corollary 2.1

For any set X , there is a monoid M and a function $p : X \rightarrow U(M)$, with U the forgetful functor, satisfying the universal property of free monoids.

Graphs can be used to describe diagrams. For any graph G , a diagram in the shape of G in the category B is a morphism $D : G \rightarrow B$ of graphs. This corresponds to a functor $D' : C_G \rightarrow B$ via the bijection $D' \mapsto D = U(D') \circ P$. This bijection

$$\mathbf{Cat}(C_G, B) \cong \mathbf{Grph}(G, U(B))$$

arises from an adjunction relation between F and U .

2.8 Quotient Categories

Proposition 2.3

Let C be a category, and let R be a function mapping each pair (a, b) in C to a binary relation $R_{a,b}$ on $C(a, b)$. Then there exists a category C/R and a functor $\pi : C \rightarrow C/R$ such that

- If $f R_{a,b} f'$ in C , then $\pi(f) = \pi(f')$;
- If $H : C \rightarrow D$ is any functor such that $f R_{a,b} f'$ implies $H(f) = H(f')$, then there is a unique functor $H' : C/R \rightarrow D$ such that $H = H' \circ \pi$ - in other words, π is universal with respect to this property. The unique functor is a bijection on objects.

Proof. R is a congruence if

- $R_{a,b}$ is an equivalence relation for all pairs;
- If $f, f' : a \rightarrow b$ have $f R_{a,b} f'$, then for all $g : a \rightarrow a'$ and all $h : b \rightarrow b'$, the compositions satisfy $(h \circ f \circ g) R_{a',b'} (h \circ f' \circ g)$ (composition preserves R).

There exists a minimal congruence R' containing R (which I refuse to prove), so we define C/R as the category where $\text{Obj}(C/R) = \text{Obj}(C)$, and $C/R(x, y) = C(x, y)/R'$. After this, anyone who has done proofs about universality of quotients can do this with their eyes closed, and it would be a waste of time to type it up instead of doing exercises. ■

Remark

Any relation that is already a congruence, such as homotopy, can be directly quotiented by without finding the minimal containing congruence. Additionally, in a similar but not equivalent vein to group presentations, if C is generated by graph G as a free category, we call C/R the group with generators G and relations R .

Chapter 3

Universals and Limits

3.1 Universal Arrows

We showed that the arrow $G \rightarrow U(C)$ from the graph into the free category formed on that graph is a *universal* arrow. This section makes precise universality in generality.

Definition 3.1: Universal Arrow

If $F : D \rightarrow C$ is a functor and $c \in C$, a universal arrow from c to F is a pair (r, u) with $r \in D$ and $u : c \rightarrow F(r)$, such that for all other pairs (d, f) with $d \in D$ and $f : c \rightarrow F(d)$, there is a unique arrow $f' : r \rightarrow d$ making the diagram

$$\begin{array}{ccc} c & \xrightarrow{u} & F(r) \\ & \searrow f & \downarrow F(f') \\ & & F(d) \end{array} \quad \begin{array}{c} r \\ \downarrow f' \\ d \end{array}$$

commute. That is, all other pairs (d, f) factor uniquely through the universal arrow under F .

Another way to view universal arrows from c to F is as initial objects in the comma category $(c \downarrow F)$: recall that objects are pairs (d, f) with $d \in D$ and arrows $f : c \rightarrow F(d)$ in C , and morphisms $(d_1, f_1) \rightarrow (d_2, f_2)$ are arrows $\varphi : d_1 \rightarrow d_2$ in D such that the diagram

$$\begin{array}{ccc} c & & \\ f_1 \downarrow & \searrow f_2 & \\ d_1 & \xrightarrow{F(\varphi)} & d_2 \end{array}$$

commutes. It then becomes clear that an initial object generates the same diagram as in definition 3.1.

An example of universal arrows arises in any situation with a free object and a forgetful functor. For the purposes of formalism, I will use the fact that free objects give rise to the free functor. Let $F : C \rightarrow D$ be a free functor with $U : D \rightarrow C$ a forgetful functor, and let $x \in C$. The free object $F(x)$ on x satisfies the universal property of free objects:

$$\begin{array}{ccc} x & \xrightarrow{i} & U(F(x)) \\ & \searrow f & \downarrow U(f') \\ & & U(z) \end{array} \quad \begin{array}{c} x \\ \downarrow f' \\ z \end{array}$$

Note that this is precisely the same as the statement that $(F(x), i)$ is a universal arrow from x to U .

Another example is with quotients. The universal property of quotients states that the quotient of a set S by a relation R satisfies the property

$$\begin{array}{ccc} S & \xrightarrow{f} & Z \\ \pi \downarrow & \nearrow f' & \\ S/R & & \end{array}$$

We can view this as a universal arrow as well. Let $H : \mathbf{Set} \rightarrow \mathbf{Set}$ be the functor mapping sets X to the subset of $\text{Hom}_{\mathbf{Set}}(S, X)$ consisting of functions $f : S \rightarrow X$ for which sRs' implies $fs = fs'$. The quotient $(S/R, \pi)$ is then immediately seen to be a universal arrow from S to H . This process works in other categories as well. This universal property is quite useful; for example in **Grp**, all isomorphism theorems and other properties of quotient groups follow easily without so much as mention of cosets.

This characterization allows for construction of universal objects which are initial with respect to some property, but the duality of universal arrows allows for the construction of terminal objects as well.

Definition 3.2: Universal Arrows, Again

Let $F : D \rightarrow C$ be a functor, with $c \in C$. A universal arrow from F to c is a pair (r, v) with $r \in D$ and $v : F(r) \rightarrow c$, which is terminal with respect to this property: for each pair (f, d) with $d \in D$ and $f : F(d) \rightarrow c$, there is a unique morphism $f' : d \rightarrow r$ such that the diagram

$$\begin{array}{ccc} F(d) & \xrightarrow{f} & c \\ \downarrow F(f') & \nearrow v & \\ F(r) & & \end{array} \quad \begin{array}{ccc} d & & \\ \downarrow f' & & \\ r & & \end{array}$$

commutes. Equivalently, a universal arrow from F to c is a terminal object in $(F \downarrow c)$.

An important example of a universal arrow in this regard is a product. Let C be a category with finite products, and let $x, y \in C$. The product is well-known to satisfy the universal property

$$\begin{array}{ccccc} & & z & & \\ & \swarrow f & \downarrow & \searrow g & \\ x & \xleftarrow{\pi_x} & x \times y & \xrightarrow{\pi_y} & y \end{array}$$

To view this as a universal arrow, let $\Delta : C \rightarrow C \times C$ be the diagonal functor

$$x \mapsto (x, x), \quad f \mapsto (f, f).$$

This is manifestly functorial. Let $r \in C$ and $p : r \rightarrow x, q : r \rightarrow y$ be morphisms in C . Then we have that $(p, q) : (r, r) \rightarrow (x, y)$. Let $(r, (p, q))$ be a universal arrow from Δ to (x, y) . We then have that the diagram

$$\begin{array}{ccc} (r, r) & \xrightarrow{(p, q)} & (x, y) \\ \downarrow (f', f') & \nearrow (f, g) & \\ (z, z) & & \end{array} \quad \begin{array}{ccc} r & & \\ \downarrow f' & & \\ z & & \end{array}$$

commutes for all $z \in C$ and for all pairs $f : z \rightarrow x$ and $g : z \rightarrow y$. Using our knowledge of product categories, the universal property of products follows immediately, and thus $(r, (p, q))$ satisfy the universal property of products.

3.2 The Yoneda Lemma

Proposition 3.1

Let $F : D \rightarrow C$ be a functor. A pair $(r, u : c \rightarrow F(r))$ is universal from c to F if and only if for each pair (d, f) , the function mapping the induced morphism f' to the composition $F(f') \circ u : c \rightarrow F(d)$ is a bijection of hom-sets

$$D(r, d) \cong C(c, F(d)).$$

This bijection is natural in d , meaning the function is a natural isomorphism. Conversely, any natural isomorphism $D(r, d) \cong C(c, F(d))$ in d is determined by a universal arrow $u : c \rightarrow F(r)$.

Proof. Universality immediately guarantees that the function is a bijection. Let f, g both have f' as the induced morphism. By commutivity of the relevant diagram, $F(f') \circ u = f$ and $F(f') \circ u = g$, and thus $f = g$. Thus, the function is injective, and surjectivity follows by definition, as for any morphism $g : c \rightarrow F(d)$, universality says there exists a morphism $g' : r \rightarrow d$ which maps to g under this function. For the converse direction, surjectivity guarantees that such a morphism f' always exists, and from this injectivity guarantees universality in much the same way.

Naturality of the function in d , which I will denote by ν , means that

$$\nu : D(r, -) \Rightarrow C(c, -) \circ F = C(c, F(-)).$$

Proof of this amounts to a proof that for all $\varphi : d \rightarrow d'$, the diagram

$$\begin{array}{ccc} D(r, d) & \xrightarrow{D(r, \varphi)} & D(r, d') \\ \nu(d) \downarrow & & \downarrow \nu(d') \\ C(r, F(d)) & \xrightarrow{C(r, F(\varphi))} & C(r, F(d')) \end{array}$$

commutes. Working elementwise, we have that the following must be equivalent:

$$\begin{array}{lcl} f' \longmapsto \varphi \circ f' \longmapsto F(\varphi \circ f) \circ u \\ f' \longmapsto F(f') \circ u \longmapsto F(\varphi) \circ F(f') \circ u \end{array}$$

This follows from functoriality of F , and thus ν is natural in d .

For the converse, let $f' : r \rightarrow d$. Naturality of ν guarantees commutivity of the diagram

$$\begin{array}{ccc} D(r, r) & \xrightarrow{\nu(r)} & C(c, F(r)) \\ D(r, f') \downarrow & & \downarrow C(c, F(f')) \\ D(r, d) & \xrightarrow{\nu(d)} & C(c, F(d)) \end{array}$$

Consider in particular the image of 1_r under both of these paths. Define $u := \nu(r)(1_r)$. We have

$$\begin{array}{ccc} 1_r & \xrightarrow{\nu(r)} & u \xrightarrow{C(c, F(f'))} F(f') \circ u \\ 1_r & \xrightarrow{D(r, f')} & f' \xrightarrow{\nu(d)} \nu(d)(f') \end{array}$$

We thus have that $\nu(d)(f') = F(f') \circ u$. Since $\nu(d)$ is a bijection for all d , it follows that f' is necessarily unique (which is equivalent here to the fact that f is unique), and thus (r, u) is universal. ■

If C and D are locally small, it follows that the functor $C(c, -) \circ F$ is naturally isomorphic to a hom-functor $D(r, -)$. This is known as a representation.

Definition 3.3: Representation

Let D be locally small. A representation of a functor $F : D \rightarrow \mathbf{Set}$ is a pair (r, ψ) with $r \in D$ and ψ a natural isomorphism

$$\psi : D(r, -) \cong F.$$

The object r is called the representing object, and the functor F is said to be *representable* if a representation exists.

Representable functors are just hom-functors up to isomorphism. They can be connected to universal arrows through the following result.

Proposition 3.2

Let $*$ denote a singleton, and let D be locally small. If $(r, u : * \rightarrow F(r))$ is universal from $*$ to $F : D \rightarrow \mathbf{Set}$, then the function $\psi : f' \mapsto F(f')(u(*)) \in F(d)$ is a representation of F , with r the representing object. Every representation of F is obtained in this way from exactly one such universal arrow.

Proof. I will begin by making clear what the proposition says. u is universal from $*$ to F , and the object obtained by evaluating $F(f') \circ u$ at $*$ is the representing object.

Since functions $f : * \rightarrow X$ are fully determined by $f(*)$, there is a bijection $\text{Hom}_{\mathbf{Set}}(*, X) \rightarrow X$ given by $f \mapsto f(*)$. This is clearly a natural isomorphism $\alpha : \text{Hom}_{\mathbf{Set}}(*, -) \cong 1_{\mathbf{Set}}$, so composing with F gives a natural isomorphism $\text{Hom}_{\mathbf{Set}}(*, F(-)) \cong F$ by horizontal composition $\alpha \circ 1_F$. By proposition 3.1, we also have $\varphi : \text{Hom}_{\mathbf{Set}}(*, F(-)) \cong C(r, -)$, and thus vertical composition $\varphi \cdot \alpha$ gives a natural isomorphism $F(-) \cong C(r, -)$. ■

Note that a universal arrow from c to $F : D \rightarrow C$ amounts to an isomorphism $D(r, d) \cong C(c, F(d))$, and hence a representation $D(r, -) \cong C(c, F(-))$, or equally a universal element of the functor. This discussion now leads us naturally to the Yoneda lemma, a result of fundamental importance in category theory.

Lemma 3.1 (Yoneda)

Let C be locally small, $F : C \rightarrow \mathbf{Set}$ be a functor, and let $c \in C$. Then there is a bijection

$$y : \text{Nat}(C(c, -), F) \cong F(c)$$

given by mapping natural transformations $\alpha : C(c, -) \Rightarrow F$ to $\alpha_c(1_c)$.

Proof. The map is well-defined; if $\alpha = \alpha'$, then clearly $\alpha_c = \alpha'_c$, and thus $\alpha_c(1_c) = \alpha'_c(1_c)$.

Let $\alpha : C(c, -) \Rightarrow F$. Then for each morphism $f : c \rightarrow d$, there is a commutative diagram

$$\begin{array}{ccc} C(c, c) & \xrightarrow{\alpha_c} & F(c) \\ \downarrow C(c, f) & & \downarrow F(f) \\ C(c, d) & \xrightarrow{\alpha_d} & F(d) \end{array}$$

Denote $\alpha_c(1_c)$ by u . Then by commutativity of the diagram, we have the equality

$$F(f)(u) = \alpha_d(f).$$

Thus, each component α_d is determined by the value of u . This shows injectivity, as if $u = u'$, then the corresponding natural transformations α, α' are necessarily equal, since for each d , for each $f : c \rightarrow d$, the value $\alpha_d(f) = F(f)(u) = F(f)(u') = \alpha'_d(f)$.

Finally, to show surjectivity, let $u \in F(c)$. Suppose there is a function α defined such that $\alpha_c(1_c) = u$, and for all $f : c \rightarrow d$, we have $\alpha_d(f) = F(f)(u)$. This makes the above diagram commute, and thus α is natural, concluding the proof. ■

Corollary 3.1

For all $r, s \in C$, there exists a unique arrow $u : s \rightarrow r$ such that each natural transformation $\alpha : C(r, -) \Rightarrow C(s, -)$ is given by $C(u, -)$.

Proof. First, I will break down the meaning of this corollary. The natural transformation $\alpha : C(r, -) \Rightarrow C(s, -)$ has components $\alpha_x : C(r, x) \rightarrow C(s, x)$ given by $f \mapsto f \circ u$.

By the Yoneda lemma,

$$\text{Nat}(C(r, -), C(s, -)) \cong C(s, r).$$

Thus, for each α , there exists a unique morphism $h : s \rightarrow r$ such that for each $f : c \rightarrow x$, we define $\alpha_x(f) = C(s, f)(h) = f \circ h$, where h is the element $\alpha_r(1_r)$ which uniquely determines α . Thus, each α_x is simply given by composing on the right with h , or thus $\alpha_x = C(x, h)$, and thus $\alpha = C(-, h)$. ■

The isomorphism y is natural in both F and c , which is done by viewing y as a natural transformation between the evaluation functor $(F, c) \mapsto F(c)$ and the functor $N(F, c) = \text{Nat}(C(c, -), F)$. This is done by viewing them both as functors $\text{Fun}(C, \mathbf{Set}) \rightarrow \mathbf{Set}$.

Lemma 3.2 (Yoneda)

With notation as previously, the bijections y give a natural transformation $y : N \Rightarrow E$.

This lemma is easily enough proven. Note that we thus have a fully faithful functor $Y : C^{\text{op}} \rightarrow \text{Fun}(C, \mathbf{Set})$ given by $r \mapsto C(r, -)$. It's dual is another functor $Y' : D \rightarrow \text{Fun}(D^{\text{op}}, \mathbf{Set})$, also fully faithful, and often known as the *Yoneda Embedding*.

If C is not locally small, these restrictions can be bypassed by instead considering the category **Ens** whose objects at least contain the hom-sets of C , and whose morphisms are functions between those hom-sets.

3.3 Coproducts and Colimits

Colimits will be introduced with a collection of special cases, which will all be universal constructions.

Let $\Delta : C \rightarrow C \times C$ be the diagonal functor $x \mapsto (x, x)$ and $f \mapsto (f, f)$. A universal arrow from an object (a, b) to Δ is a *coproduct diagram*. This universal arrow consists of an object $c \in C$, and arrows $i_a : a \rightarrow c$ and $i_b : b \rightarrow c$ in C ; that is, an arrow $(i_a, i_b) : (a, b) \rightarrow \Delta(c)$; satisfying the universal property that for any object $z \in C$ and any morphism $(f, g) : (a, b) \rightarrow (x, y)$, there exists a unique arrow $h : c \rightarrow z$ in C making the diagram

$$\begin{array}{ccc} (a, b) & \xrightarrow{(i_a, i_b)} & (c, c) \\ & \searrow (f, g) & \downarrow (h, h) \\ & & (z, z) \end{array} \quad \begin{array}{c} c \\ \downarrow h \\ z \end{array}$$

commute. We call the object c the *coproduct* of a and b , and write it as $a \amalg b$. We can also write the universal property as a diagram in C rather than in $C \times C$:

$$\begin{array}{ccccc} a & \xrightarrow{i_a} & a \amalg b & \xleftarrow{i_b} & b \\ & \searrow f & \downarrow h & \swarrow g & \\ & & z & & \end{array}$$

This gives us the familiar universal property of coproducts. We also sometimes write the morphism h as $f \amalg g$. If the coproducts exists, it are necessarily unique up to isomorphism, and thus the assignment $(f, g) \mapsto f \amalg g$ becomes a bijection

$$C(a, z) \times C(b, z) \cong C(a \amalg b, z),$$

which is natural in z . If all coproducts exist in C , then $\amalg$ becomes a bifunctor $C \times C \rightarrow C$. In a preorder, the coproduct of two elements a, b is any least upper bound $\inf(a, b)$ in the preorder.

Infinite coproducts are similar. We can view a finite coproduct as a universal arrow to the diagonal functor into $C \times C = C^2$. Given an infinite coproduct indexed over a set X , we can take X to define a discrete category and consider a diagonal functor to the functor category C^X , which can be viewed as the category of X -indexed families of objects in C . The X -ary diagonal functor $\Delta : C \rightarrow C^X$ maps objects c to the constant functor $x \mapsto c$ for all $x \in X$, which is viewed as the constant family $c_x = c$ for all $x \in X$. The action of the constant functor on morphisms is to map $f \mapsto 1_c$ for all f , and thus we can define constant natural transformations $f : c \rightarrow d$ where each component $f_x = f$. This is the action of Δ on morphisms, and we can view them as X -indexed families of morphisms. A coproduct

$$c = \coprod_{x \in X} c_x$$

is then a universal arrow $(c, \{i_x\}_{x \in X})$ from $\{c_x\}_{x \in X}$ to Δ . The universal property gives that the assignment $f \mapsto \{f \circ i_x\}_{x \in X}$ is a bijection

$$C\left(\coprod_{x \in X} c_x, z\right) \cong \prod_{x \in X} C(c_x, z),$$

natural in z .

If the entries in a coproduct are constant; that is $c_x = c$ for all x , then we call it a *copower* and write it as $X \cdot c$, so that

$$C(X \cdot c, z) \cong C(c, z)^X.$$

For example, in **Set**, the copower is the cartesian product $X \cdot Y = X \times Y$.

Coequalizers are a generalization of cokernels in categories where zero objects may not exist. Given a pair $f, g : a \rightarrow b$ of objects, a coequalizer $\text{coeq}(a, b)$ of a and b is a pair $(e, u : b \rightarrow e)$ such that $uf = ug$,

and u is universal with respect to this property:

$$\begin{array}{ccccc} a & \xrightarrow[f]{g} & b & \xrightarrow{u} & e \\ & & \searrow h & & \downarrow h' \\ & & & & c, \end{array}$$

which commutes with the standard rules (meaning it is not implied that $f = g$).

We can view coequalizers as universal arrows as follows. Let $\Downarrow$ denote the category

$$1 \xrightarrow[f]{g} 2.$$

A functor $\Downarrow \rightarrow C$ is then a choice of two elements, and two parallel arrows. Morphisms in the category $C^{\Downarrow}$ (natural transformations between these functors) then simply become pairs (h, k) which commute the two diagrams

$$\begin{array}{ccc} a & \xrightarrow[f]{g} & b \\ \downarrow h & & \downarrow k \\ a' & \xrightarrow[f']{g'} & b' \end{array}$$

where the images of the two implicit functors are as implied.

Define the $\Downarrow$ -ary diagonal functor $\Delta : C \rightarrow C^{\Downarrow}$ once again as mapping objects to the constant functor, and morphisms to the constant natural transformations between these functors:

$$\begin{array}{ccc} c & \xrightarrow[1]{1} & c \\ \downarrow f & \mapsto & \downarrow f \\ c' & \xrightarrow[1]{1} & c' \end{array}$$

Now return to the pair $f, g : a \rightarrow b$. An arrow h with $hf = hg$ is the same as an arrow $(hf = hg, h) : (f, g) \rightarrow (1_c, 1_c)$ in the functor category:

$$\begin{array}{ccc} a & \xrightarrow[f]{g} & b \\ hf \downarrow & & \downarrow h \\ c & \xrightarrow[1]{1} & c \end{array}$$

A coequalizer is just universal with respect to this, so a coequalizer is simply a universal arrow (e, u) from the functor identifying (f, g) to the functor Δ .

Given a pair of arrows $f, g : a \rightarrow b, c$ with common domain, a pushout of (f, g) is a triple (u, v, r) , which is universal with respect to commutative squares:

$$\begin{array}{ccccc} a & \xrightarrow{f} & b & & \\ \downarrow g & & \downarrow u & \searrow h & \\ c & \xrightarrow{v} & r & \xrightarrow{\quad} & z \\ & \searrow k & & \nearrow & \end{array}$$

We usually write $b \amalg_a c$ or $b \amalg_{(f,g)} c$ for r , and also call it the *fibred sum*. Let $\cdot \leftarrow \cdot \rightarrow \cdot$ be a category, and let the diagonal functor be exactly what intuition dictates. The pushout is then a universal arrow from (f, g) to Δ .

All of these constructions involve a universal arrow to some diagonal functor. Now is time to define colimits in generality. For any categories C, J , let $\Delta : C \rightarrow C^J$ be the diagonal functor, where $\Delta(c)$ is the constant functor $j \mapsto c$ for all $j \in J$, and $f : i \rightarrow j$ maps to $1_c : c \rightarrow c$. This allows us to define, for all $g : c \rightarrow d$ in C , the constant natural transformation $\Delta(g) : \Delta(c) \Rightarrow \Delta(d)$, where each component $\Delta(g)_j = g$.

Definition 3.4: Colimit

With notation as above, the colimit of a functor $F : J \rightarrow C$ is a universal arrow (r, u) from F to Δ , usually written as $r = \varinjlim F$.

Recall that this implies u is a natural transformation from F to $\Delta(r)$, such that for all $c \in C$ and all natural transformations $\alpha : F \Rightarrow \Delta(c)$, the diagram

$$\begin{array}{ccc} F & \xrightarrow{u} & \Delta(r) \\ & \searrow \alpha & \downarrow \Delta(\alpha') \\ & & \Delta(c) \end{array}$$

commutes.

Looking only at the fact that there is a natural transformation $u : F \Rightarrow \Delta(r)$, we have that for any morphism $f : i \rightarrow j$ in J , the diagram

$$\begin{array}{ccc} F(i) & \xrightarrow{F(f)} & F(j) \\ u_i \downarrow & & \downarrow u_j \\ \Delta(r)(i) & \xrightarrow{\Delta(r)(f)} & \Delta(r)(j) \end{array}$$

commutes. Of course, $\Delta(r)(-) = r$ and $\Delta(r)(f) = 1_r$, so this reduces to the diagram

$$\begin{array}{ccc} F(i) & \xrightarrow{F(f)} & F(j) \\ u_i \downarrow & & \downarrow u_j \\ r & \xlongequal{\quad} & r \end{array}$$

This is typically rewritten as

$$\begin{array}{ccc} F(i) & \xrightarrow{F(f)} & F(j) \\ & \searrow u_i & \swarrow u_j \\ & r & \end{array}$$

commuting for all $f : i \rightarrow j$ in J . Such an object r , together with a natural transformation u , is known as a *co-cone* for F . The universality of r and u means that for any other co-cone (c, α) for F , the diagram

$$\begin{array}{ccc} F(i) & \xrightarrow{F(f)} & F(j) \\ & \searrow u_i & \swarrow u_j \\ & r = \varinjlim F & \\ & \downarrow \alpha' & \\ & c & \end{array}$$

α_i (curved arrow from $F(i)$ to c) α_j (curved arrow from $F(j)$ to c)

commutes for any $f : i \rightarrow j$ in J .

3.4 Products and Limits

Definition 3.5: Limit

Let $F : J \rightarrow C$ be a functor. A limit of F is a universal arrow $(\varprojlim F, u)$ from the diagonal functor $\Delta : C \rightarrow C^J$ to F . Equivalently, it is a *cone* which is universal with respect to cones, where a cone is the dual of a co-cone.

More explicitly, a cone is a pair $(r, u : \Delta(r) \Rightarrow F)$, meaning that for all $f : i \rightarrow j$ in J , the diagram

$$\begin{array}{ccc} & r & \\ u_i \swarrow & & \searrow u_j \\ F(i) & \xrightarrow{F(f)} & F(j) \end{array}$$

commutes. As with any universals, if they exist, limits, colimits, and the limiting (co-)cones are uniquely determined up to isomorphism by F .

When limits exist, there are natural isomorphisms

$$C(c, \varprojlim F) \cong \text{Nat}(\Delta(c), F) = \text{Cone}(c, F);$$

$$\text{Cone}(F, c) = \text{Nat}(F, \Delta(c)) \cong C(\varinjlim F, c).$$

Examples of limits include duals to all the constructions mentioned in section 3.3, and more that were not mentioned. Examples include products, kernels, pullbacks, and equalizers.

3.5 Categories with Finite Products

A category has *finite products* if for all finite collections of objects (and by induction, all pairs of objects), there exists a product diagram with a product object and projections. In particular, C has a product of *no* objects, which would be the limit of the null functor, and thus simply a terminal object. Define the diagonal map $\delta_c : c \rightarrow c \times c$ as the induced morphism in the product diagram $1_c \times 1_c$; meaning $\pi_1 \delta_c = \pi_1 1_c = 1_c$.

Proposition 3.3

If a category C has a terminal object t and a product diagram for any pair of objects, then it has finite products. The diagrams provide a bifunctor $\times : C \times C \rightarrow C$. For any $a, b, c \in C$, there is an isomorphism $\alpha_{a,b,c} : a \times (b \times c) \rightarrow (a \times b) \times c$ called the *associator*, natural in a, b, c . Equivalently, this is a natural isomorphism $\times \circ (1_C \otimes \times) \Rightarrow \times \circ (\times \otimes 1_C)$. I have here denoted by $\otimes$ the product of functors, simply for clarity of notation. Additionally, for all $a \in C$ there are natural isomorphisms

$$\lambda_a : t \times a \cong a, \quad \rho_a : a \times t \cong a$$

called the *left-* and *right-unitors*, respectively. These are equivalently viewed as natural isomorphisms

$$\lambda : t \times - \Rightarrow 1_C, \quad \rho : - \times t \Rightarrow 1_C.$$

Proof. Existence of products is immediate by induction, and the existence of the morphisms is easily shown via a short diagram chase. I decline to include the proof in these notes to avoid drawing many

diagrams. ■

If a category C has both finite products and coproducts, then the arrow

$$\coprod_{i \in [m]} a_i \rightarrow \prod_{j \in [n]} b_j$$

is determined uniquely by an $m \times n$ matrix of arrows $f_{jk} = p_k f i_j : a_j \rightarrow b_k$, with j ranging over $[m]$ and k ranging over $[n]$. In categories of finite dimensional vector spaces, coproducts and products coincide, and thus this is a usual matrix for a change of basis.

More generally, if C has a zero object 0 , then there is a *canonical* arrow

$$\coprod_i a_i \rightarrow \prod_i a_i.$$

3.6 Groups in Categories

Given a category with finite products with t the terminal object, a monoid (this definition will be upgraded later) is a triple (c, μ, η) with $\mu : c \times c \rightarrow c$ and $\eta : t \rightarrow c$, such that the diagrams

$$\begin{array}{ccccc} c \times (c \times c) & \xrightarrow{\alpha} & (c \times c) \times c & \xrightarrow{\mu \times 1} & c \times c \\ \downarrow 1 \times \mu & & & & \downarrow \mu \\ c \times c & \xrightarrow{\mu} & c & & \\ & & & & \\ t \times c & \xrightarrow{\eta \times 1} & c \times c & \xleftarrow{1 \times \eta} & c \times t \\ & \searrow \lambda & \downarrow \mu & \swarrow \rho & \\ & & c & & \end{array}$$

commute. Here, α is the associator, λ is the left unitor, and ρ is the right unitor, as given in proposition 3.3. By viewing μ as multiplication in the monoid, and by viewing η as the unitor morphism taking the place of the identity element of a monoid, this definition makes perfect sense. Taking δ_c to once again be the diagonal morphism $c \rightarrow c \times c$, given by the product $1_c \times 1_c$, we can extend this definition to encompass that of a group by introducing an “inversion” morphism $i : c \rightarrow c$ (the book uses ζ , I think thats stupid).

Definition 3.6: Group Object

A group object in a category C with finite products is a pair (c, μ, η, i) such that (c, μ, η) form a monoid with respect to the induced monoidal structure by $\times$ (a monoid as described above), and such that the diagram

$$\begin{array}{ccccc} c & \xrightarrow{\delta_c} & c \times c & \xrightarrow{1 \times i} & c \times c \\ \downarrow \text{dashed} & & & & \downarrow \mu \\ t & \xrightarrow{\eta} & c & & c \end{array}$$

commutes. We call μ multiplication, η the unitor, and i inversion.

A group object in **Set** is easily shown to simply be a group, if we take μ is the standard multiplication, the image $\eta(*)$ to be the identity (since in **Set** $t = *$ is the singleton), and i to be the map $c \mapsto c$. It is easily shown that given a group object, these definitions satisfy the group axioms.

It is easily shown that given any group G and any set X , the set $G^X = \text{Hom}_{\mathbf{Set}}(X, G)$ has a group structure given by pointwise multiplication. This generalizes via the Yoneda lemma.

Proposition 3.4

Let C be a category with finite products. Then an object c is a group object if and only if the hom-functor $\text{Hom}(-, c)$ is a group in the functor category $\mathbf{Set}^{C^{\text{op}}}$.

Proof. Recall that by universality of product, for any $f : z \rightarrow a \times b$, by proposition 3.1 there is a natural bijection $C(z, a) \times C(z, b) \cong C(z, a \times b)$ given by $(\pi_a f, \pi_b f) \mapsto h$, and vice versa. We thus have that multiplication $\mu : c \times c \rightarrow c$ induces a multiplication $\bar{\mu}$ on $C(-, c) \times C(-, c)$ by the composition

$$C(-, c) \times C(-, c) \longrightarrow C(-, c \times c) \xrightarrow{C(-, \mu)} C(-, c),$$

which for any pair $(f : a \rightarrow c, g : b \rightarrow c)$ maps

$$(f, g) \longmapsto f \times g \longmapsto \mu \circ (f \times g).$$

I don't feel like finishing this proof. ■

3.7 Colimits of Representable Functors

Theorem 3.1

Let $F : C \rightarrow \mathbf{Set}$ with D small. Then F can be represented canonically as a colimit of a diagram of representable functors $C \rightarrow \mathbf{Set}$.

Proof. Let $J = (1 \downarrow F)$, where we remark that 1 is a singleton (so we are changing notation from $*$ to 1). Since morphisms $1 \rightarrow z$ identify elements of z , the category J is thus the category of elements of F , meaning objects are pairs (d, x) for $x \in F(d)$ and $d \in C$, and morphisms $(d, x) \rightarrow (d', x')$ are arrows $f : d \rightarrow d'$ such that $F(f)(x) = x'$.

Let $M : J^{\text{op}} \rightarrow \mathbf{Set}^C$ be the functor given by $(x, d) \mapsto C(d, -)$; and each arrow $f : d \rightarrow d'$ mapped to the precomposition $D(f, -) : g \mapsto g \circ f$. We will write $D(f, -)$ as f^* for the purposes of this proof. This is basically the Yoneda embedding, but applied over the category of elements of F , so we would expect the proof may fall out of the Yoneda lemma. Indeed, the inverse Yoneda isomorphism

$$y_d^{-1} : F(d) \rightarrow \text{Nat}(C(d, -), F), \quad u \mapsto \alpha^u$$

where $\alpha_c^u : D(d, c) \rightarrow F(c)$ is the function $g \mapsto F(g)(c)$. We claim this yields a universal co-cone for M over $\mathbf{Set}^D$ with F as the colimit, given by maps $\mu : M(d, x) \rightarrow F$ given by $\mu_d^x = y_d^{-1}(x)$.

To show this, let $F(f) : (d, x) \rightarrow (d', x')$, and thus the image under M is

$$C(d, -) \xrightarrow{f^*} C(d', -).$$

Naturality of the Yoneda isomorphism in d gives that

$$\begin{array}{ccc} C(d, -) & \xrightarrow{f^*} & C(d', -) \\ y_d^{-1} \downarrow & & \downarrow y_{d'}^{-1} \\ F & \xlongequal{\quad} & F \end{array}$$

| I don't think this works but I give up this proof is aids. ■

Contravariant functors $F : C^{\text{op}} \rightarrow \mathbf{Set}$ are often called *presheaves*; the functor category $\mathbf{Set}^{C^{\text{op}}}$ is often written as $\hat{C}$ to indicate this. The intuition comes from the topological definition of a presheaf falling out when applied to the poset category of open sets of a topological space.

Chapter 4

Adjoints

4.1 Adjunctions

The following notation appears to be fairly dated, but I will use it over the course of this book since Mac Lane uses it.

Definition 4.1: Adjunction

Let C, D be categories. An adjunction from C to D is a triple $(F, G, \varphi) : D \rightarrow C$ where F and G are functors

$$C \begin{array}{c} \xrightarrow{F} \\ \xleftarrow{G} \end{array} D,$$

and φ is a function which maps pairs $(c, d) \in C \times D$ to an isomorphism

$$\varphi_{(c,d)} : D(F(c), d) \cong C(c, G(d)),$$

which is natural in c and d .

The naturality condition in φ asserts that φ is a natural isomorphism between the compositions of bifunctors

$$\varphi : D(-, -) \circ (F^{\text{op}} \times 1_D) \Rightarrow C(-, -) \circ (1_{C^{\text{op}}} \times G),$$

where we recall that $F^{\text{op}} : C^{\text{op}} \rightarrow D^{\text{op}}$. This is necessary due to how the hom functors work, as we well know. More concisely,

$$\varphi : D(F(-), -) \Rightarrow C(-, G(-)).$$

The naturality condition thus asserts that for all $f : c \rightarrow c'$ and $g : d \rightarrow d'$, the diagram

$$\begin{array}{ccc} D(F(c), d) & \xrightarrow{\varphi_{(c,d)}} & C(c, G(d)) \\ \downarrow D(F(f), g) & & \downarrow C(f, G(g)) \\ D(F(c'), d') & \xrightarrow{\varphi_{(c',d')}} & C(c', G(d')) \end{array}$$

commutes. This is equivalently viewed as the book presents it, where we have the commutivity of 2 diagrams, one only considering f and one only considering g . As is, the morphism $C(F(f), g)$ is the map $\varphi \mapsto g \circ \varphi \circ F(f)$, and $D(f, G(g))$ is similar.

With notation as above, we say F is *left-adjoint* to G , and G is *right-adjoint* to F , since F appears on the left and G appears on the right in the hom-sets. We also call F a *left-adjoint functor*, and G a *right-adjoint functor*. We write $F \dashv G$.

Each adjunction gives rise to universal arrows. Let $d = F(c)$. Then the natural isomorphism $\varphi_{(c,d)}$ gives

$$D(F(c), F(c)) \cong C(c, G(F(c))).$$

We then have $1_{F(c)} \in D(F(c), F(c))$, so define $\eta_c = 1_{F(c)}$. By proposition 3.1, we thus have that $(\eta_c, F(c))$ is universal from c to G (since each map $c \rightarrow G(d)$ is in bijection with a map $F(c) \rightarrow d$), and thus we have an assignment $c \mapsto \eta_c$ to universal arrows *for all* $c \in C$. Moreover, this assignment $c \mapsto \eta_c$ is natural $1_C \Rightarrow G \circ F$, since naturality of φ guarantees that the diagram

$$\begin{array}{ccc} c' & \xrightarrow{\eta_{c'}} & G(F(c')) \\ \downarrow h & & \downarrow G(F(h)) \\ c & \xrightarrow{\eta_c} & G(F(c)) \end{array}$$

since $\eta_c = \varphi(1_{F(c)})$. By proposition 3.1, we have that the bijection φ can be given by

$$\varphi_{(c,d)}(f) = G(f) \circ \eta_c.$$

I think it's good to show this explicitly here. By naturality of φ , for each $f : F(c) \rightarrow d$, the diagram

$$\begin{array}{ccc} D(F(c), F(c)) & \xrightarrow{\varphi} & C(c, GF(c)) \\ \downarrow D(F(c), f) & & \downarrow C(c, G(f)) \\ D(F(c), d) & \xrightarrow{\varphi} & C(c, G(d)) \end{array}$$

commutes. Chasing the arrow $1_{F(c)}$ around the diagram gives

$$G(f) \circ \varphi(1_{F(c)}) = \varphi(f \circ D(F(c), 1_{F(c)}).$$

By functorality, the left simplifies to $\varphi(f \circ 1_{F(c)}) = \varphi(f)$, and the right is defined as $G(f) \circ \eta_c$. This gives us this equality.

This all follows as well for F . Consider the bijection with $c = G(d)$:

$$\varphi : D(FG(d), d) \cong C(G(d), G(d)).$$

Define by ε_d the image $\varphi^{-1}(1_{G(d)})$, and by much the same arguments, we have that $(\varepsilon_d, G(d))$ is universal from F to d , which is essentially the same as the statement that

$$\varphi^{-1}(f) = \varepsilon_d \circ F(f), \quad f : c \rightarrow G(d).$$

Finally, the map $d \mapsto \varepsilon_d$ is natural $\varepsilon : FG \Rightarrow 1_D$.

Consider the map $1_{G(d)} : G(d) \rightarrow G(d)$. We have by the previous equality that

$$\varphi^{-1}(1_{G(d)}) = \varepsilon \circ \varepsilon_d \circ F(1_{G(d)}) = \varepsilon_d \circ 1_{FG(d)} = \varepsilon_d.$$

We thus have

$$1_{G(d)} = (\varphi \circ \varphi^{-1})(1_{G(d)}) = \varphi(\varepsilon_d).$$

Since $\varepsilon_d : FG(d) \rightarrow d$, we have that

$$\varphi(\varepsilon_d) = G(\varepsilon_d) \circ \eta_{G(d)}.$$

We typically write this condition diagrammatically as the commutivity of the diagram

$$\begin{array}{ccccc} & & 1_G & & \\ & \searrow & & \nearrow & \\ G & \xrightarrow{\eta_G} & GFG & \xrightarrow{G\varepsilon} & G. \end{array}$$

By duality, the entirety of the following statement follows.

Theorem 4.1

An adjunction $(F, G, \varphi) : D \leftarrow C$ determines

1. A natural transformation $\eta : 1_C \Rightarrow GF$ such that each component η_c is universal from c to G , and such that for all $f : F(c) \rightarrow d$,

$$\varphi(f) = G(f) \circ \eta_c.$$

2. A natural transformation $\eta : FG \Rightarrow 1_D$ such that each component ε_d is universal from F to d , and such that for all $g : c \rightarrow G(d)$,

$$\varphi^{-1}(g) = \varepsilon_d \circ F(g).$$

3. The following composites are the identity natural transformations:

$$G \xrightarrow{\eta G} GFG \xrightarrow{G\varepsilon} G$$

$$F \xrightarrow{F\eta} FGF \xrightarrow{\varepsilon F} F$$

We call η the *unit* and ε the *counit* of the adjunction. The results of this theorem are in fact shown to determine adjunctions in various ways. This is summarized by the following theorem.

Theorem 4.2

Each adjunction $(F, G, \varphi) : C \leftarrow D$ is completely determined by any of the following.

1. Functors F, G and a natural transformation $\eta : 1_C \Rightarrow GF$ such that each $\eta_x : c \rightarrow GF(c)$ is universal from c to G ;
2. The functor $G : D \rightarrow C$, an object function $F_0 : X \rightarrow A$, and a universal arrow $\eta_c : c \rightarrow GF_0(c)$ for all $c \in C$;
3. Functors F, G and a natural transformation $\varepsilon : FG \Rightarrow 1_D$ such that each $\varepsilon_d : FG(d) \rightarrow d$ is universal from F to d ;
4. The functor $F : C \rightarrow D$, an object function $G_0 : D \rightarrow C$, and a universal arrow $\varepsilon_d : FG(d) \rightarrow d$ for each $d \in D$;
5. Functors F, G and natural transformations $\eta : 1_C \Rightarrow GF$ and $\varepsilon : FG \Rightarrow 1_D$ such that the composites $G\varepsilon \circ \eta G$ and $\varepsilon F \circ F\eta$ are the identity natural transformations.

Proof. Assume (1). Choose $c \in C$ and $d \in D$, and define the bijection $\varphi_{(c,d)} : D(F(c), d) \rightarrow C(c, G(d))$ via the previous identification

$$\varphi_{(c,d)}(f) = G(f) \circ \eta_c.$$

Each pair $(\eta_c, F(c))$ is universal from c to G , so proposition 3.1 gives the bijections

$$D(F(c), d) \cong C(c, G(d))$$

for any $d \in D$, natural in d . For the converse, let $f : c \rightarrow c'$, and consider the diagram

$$\begin{array}{ccc} D(F(c), d) & \xrightarrow{\varphi} & C(c, G(d)) \\ D(F(c), f) \downarrow & & \downarrow C(c, G(f)) \\ D(F(c'), d) & \xrightarrow{\varphi} & C(c', G(d)) \end{array}$$

The definition of φ gives

$$f \circ G(g) \circ \eta_c = GF(f) \circ G(g) \circ \eta_c$$

for any $g : F(c) \rightarrow d$. idk chase soemthign in the diagram ig?

Assume (2), we show (1). We simply show there exists a unique way to make F_0 into a functor for which η will be natural. Let $h : c \rightarrow c'$; universality of η_c guarentees there exists a unique arrow $h' : F_0(c) \rightarrow F_0(c')$ making the diagram

$$\begin{array}{ccc} c & \xrightarrow{\eta_c} & GF_0(c) \\ h \downarrow & & \downarrow G(h') \\ c' & \xrightarrow{\eta_{c'}} & GF_0(c') \end{array}$$

commute. Thus, η is natural if we define $F(h) = h'$, and this is clearly functorial by universality.

(3,4) are dual to (1,2).

Assume (5); define φ by

$$\begin{aligned} \varphi(f) &= G(f) \circ \eta_x, & f : F(c) \rightarrow d, \\ \varphi^{-1}(g) &= \varepsilon_d \circ F(g), & g : c \rightarrow G(d). \end{aligned}$$

Since G is a functor and η is natural,

$$(\varphi \circ \varphi^{-1})(g) = \varphi(\varepsilon_d \circ F(g)) = G(\varepsilon_d) \circ GF(g) \circ \eta_c = G(\varepsilon_d) \circ \eta_{G(d)} \circ g = g.$$

Therefore, φ is a bijection of hom-sets. It is moreover clearly natural. ■

This theorem is useful, since it allows us to construct adjunctions any time we have a universal arrow from/to every object in a category to/from the same functor. For example, let C have finite products. Then there is a universal arrow from $c \in C$ to Δ for all $c \in C$, and thus the function $(a, b) \mapsto a \times b$ is in fact a functor $C \times C \rightarrow C$ by the previous theorem, and $\Delta \dashv \times$. We thus have

$$(C \times C)(\Delta(c), (a, b)) \cong C(c, a \times b) \implies C(c, a) \times C(c, b) \cong C(c, a \times b).$$

Similarly, a category with coproducts admits a coproduct functor which is left-adjoint to Δ . Any limit that can be constructed and always exists is adjoint to some diagonal functor, and it should be clear that proving universality is a simple way of showing the existence of an adjoint.

The final condition makes discussion of adjoints easy, since manipulation of these simple identities, known as the *triangle identities*, is very simple in comparison to some of the alternatives.

Corollary 4.1

Left-adjoints F of a functor G are unique up to isomorphism.

Proof. Adjunctions $F \dashv G$ and $F' \dashv$ give universal arrows $x \rightarrow GF(x)$ and $x \rightarrow GF(x')$, which must be isomorphic by universality. The proof from her is obvious. ■

Corollary 4.2

A functor $G : D \rightarrow C$ has a left adjoint if and only if for all $c \in C$, the functor $C(c, G(d))$ is representable as a functor of $d \in D$.

Proof. This is a restatement of (2) from theorem 4.2. A representation would in this case be a natural isomorphism

$$\varphi : C(c, G(d)) \cong D(F_0(x), a)$$

for an object function F_0 . ■

Theorem 4.3

If $G : D \rightarrow C$ is additive between Ab -categories, and admits a left-adjoint $F : C \rightarrow D$, then F is additive, and the adjunction is an isomorphism of abelian groups.

Proof. If η is the unit of the adjunction, then $\varphi(f) = G(f) \circ \eta_c$ for any $f : F(c) \rightarrow d$. Additivity gives

$$\varphi(f + f') = G(f + f') \circ \eta_c = G(f)\eta_c + G(f')\eta_c = \varphi(f) + \varphi(f').$$

Next, we have

$$GF(g + g') \circ p\eta_c = \eta_{c'}(g + g')$$

by naturality of η , and thus additivity of F follows easily from that of G . ■

4.2 Examples of Adjoints

<i>Forgetful Functor</i>	<i>Left Adjoint F</i>	<i>Unit of Adjunction</i>
$U : R\text{-Mod} \rightarrow \mathbf{Set}$	Free R -modules	insertion of generators $j : X \rightarrow UF(X)$
$U : \mathbf{Cat} \rightarrow \mathbf{Grph}$	Free category	insertion of generators $G \rightarrow U(C(G))$
$U : \mathbf{Grp} \rightarrow \mathbf{Set}$	Free groups	insertion of generators
$U : \mathbf{Ab} \rightarrow \mathbf{Set}$	Free abelian groups	insertion of generators
$U : \mathbf{Ab} \rightarrow \mathbf{Grp}$	Quotient by the center $G \mapsto G/Z(G)$	projection onto the quotient $\pi : G \rightarrow G/Z(G)$
$U : R\text{-Mod} \rightarrow \mathbf{Ab}$	$A \mapsto R \otimes A$	$A \rightarrow U(R \otimes A)$ by $a \mapsto 1 \otimes a$

A completion of this table, and a similar table for (co)limits, can be found on pages 87/88. I do not feel the need to copy down reference material, I would rather compute it myself.

4.3 Reflexive Subcategories

For many forgetful functors, the counit $\eta : FU \Rightarrow 1_A$ maps $a \mapsto \eta_a : F(U(a)) \rightarrow a$, which turns out to also be an epimorphism. This is a fact, and the proof requires the following lemma.

Lemma 4.1

Let $f^* : A(a, -) \Rightarrow A(b, -)$ be the natural transformation induced by an arrow $f : b \rightarrow a$ in A . Then f^* is monic if and only if f is epi, and f^* is epi if and only if f is a *split* monic.

Proof. By the Yoneda lemma, $f^* \mapsto f$ is a natural bijection, and by a previous exercise a natural transformation is monic/epi if and only if every component is monic/epi.

For $f \in A(a, c)$, by definition $f^*(h) = h \circ f$. Since f^* is a function, assume it is injective, since that is equivalent to monic. We thus have $f^*(h) = f^*(k)$ implies $h = k$, and rewriting this as above gives

$$h \circ f = k \circ f \implies h = k.$$

Thus, f is epi, and the converse is the same. This proves the first statement, so now assume f^* is epi. It is thus true that each component is epi, and thus the map $f^* : A(a, b) \rightarrow A(b, b)$ is surjective. There thus exists $h_0 : a \rightarrow b$ such that $f^*(h_0) = 1_b$. Expanding this gives $h_0 \circ f = 1_b$, and f has a left-inverse. The converse is trivial. ■

Theorem 4.4

Let $(F, G, \eta, \varepsilon) : X \rightarrow A$ be an adjunction.

1. G is faithful if and only if every component ε_a is epi;
2. G is full if and only if every component ε_a is a split monic.

Hence G is fully faithful if and only if each ε_a is an isomorphism $FG(a) \cong a$.

Proof. Note that the arrow function $G_{a,c} : A(a, c) \rightarrow X(G(a), G(c))$ turns into a natural transformation by the composition

$$A(a, c) \xrightarrow{G_{a,c}} X(G(a), G(c)) \xrightarrow{\varphi} A(FG(a), c).$$

By the Yoneda lemma, this component of the natural transformation is fully determined by $G(1_a)$ (easily seen by setting $c = a$). This is precisely η_a . Since φ is an isomorphism, this composition is epi if and only if $G_{a,c}$ is epi, and vice versa for mono. This is viewed again as injective vs surjective, hence faithful vs full. Note that we can view this as precomposition by $G_{a,c}$, and thus the proof follows from lemma 4.1. ■

Definition 4.2: Reflective Subcategory

A subcategory A of B is *reflective* if the inclusion functor $i : A \hookrightarrow B$ admits a left-adjoint $F : B \rightarrow A$, called the *reflector*. The adjunction $(F, i, \eta, \varepsilon)$ is called a *reflection*. Dually, A is *coreflective* if i has a right-adjoint.

Since the inclusion functor is really an identity, this can be described more succinctly as a functor $R = KF : B \rightarrow B$ such that we have a bijection

$$A(R(b), a) \cong B(b, a).$$

Since the inclusion is always faithful, the counit ε is *always* epi. For example, **Ab** is reflective in **Grp**.

4.4 Equivalence of Categories

Categories are isomorphic iff there exist invertible functors between them. In this case, there is an adjunction $(F, G, 1_F, 1_G)$ with $F \circ G = 1$ and $G \circ F = 1$. This is a very strict notion, and it is in fact often more useful to work in a more general notion.

Definition 4.3: Equivalence of Categories

A functor $F : C \rightarrow D$ is an *equivalence of categories* when there exists a functor $G : D \rightarrow C$ with $FG \cong 1_D$ and $GF \cong 1_C$; that is, they are invertible up to isomorphism. G is also an equivalence of categories here.

An example arises in the following useful definition.

Definition 4.4: Skeleton

A *skeleton* of a category C is any full subcategory A such that each object in C is isomorphic to *exactly one* object in A ; in other words, A contains precisely one element from all equivalence classes of C .

It is easily shown that given a skeleton $A \subseteq C$, the inclusion is an equivalence of categories. For each $c \in C$, let $a_c \in A$ have $a_c \cong c$. Define $F : C \rightarrow A$ by $c \mapsto a_c$. Since they are isomorphic, by the Yoneda lemma they have a bijection of hom-sets, so there is an apparent way to define action on morphisms as well, although I do not feel like typing up the diagram. Then clearly we have that for any $a \in A$, $F(i(a)) = a \cong a$, and for any $c \in C$, $i(F(c)) = a_c \cong c$.

An *adjoint equivalence* of categories is an adjunction $(F, G, \eta, \varepsilon)$ in which both the unit and counit are natural isomorphisms. It's not hard to show by pulling out the natural isomorphism in adjunctions that you then get the composition $G\eta^{-1} \circ \varepsilon^{-1}G = 1$, and similarly for F , giving a reverse adjunction $(G, F, \varepsilon^{-1}, \eta^{-1})$. Thus, F is both the left- and right-adjoint of G simultaneously, and vice versa.

Theorem 4.5

Let $G : C \rightarrow A$. The following are equivalent:

1. G is an equivalence of categories;
2. G is part of an adjoint equivalence;
3. G is fully faithful, and each $c \in C$ is isomorphic to $F(a)$ for some $a \in A$;

Proof. (2) implies (1) by definition, since the unit $\eta : 1_C \Rightarrow GF$ and $\varepsilon : FG \Rightarrow 1_A$ need to be isomorphisms.

(1) implies (3) fairly easily; since $GF \cong 1_C$ by some natural transformation α , we have that each $GF(c) \cong c$ by α for all $c \in C$. Set $a = F(c) \in A$. To show G is fully faithful, α gives a commutative square

$$\begin{array}{ccc} FG(a) & \xrightarrow{\alpha_a} & a \\ \downarrow FG(f) & & \downarrow f \\ FG(a') & \xrightarrow{\alpha_{a'}} & a' \end{array}$$

for all $f : a \rightarrow a'$ in A ; hence we have

$$f \circ \alpha_a = \alpha_{a'} \circ FG(f) \implies f = \alpha_{a'} \circ FG(f) \circ \alpha_{a'}^{-1}.$$

Apparently this shows G is faithful, I don't fully see it but sure. This proves T is faithful by symmetry. By commutivity of the above square and the fact that G is faithful, the fact that G is full follows very easily.

To show (3) implies (2), we construct from G a left-adjoint F . By theorem ??, we have that ε is a natural isomorphism. I'll maybe finish this tomorrow. ■

An interesting result is that any full subcategory $A \subseteq C$ satisfying $\forall c \in C \exists a \in A$ such that $c \cong a$ is reflective. This is unsurprising; it amounts to simply saying any full subcategory containing a skeleton is equivalent to C .